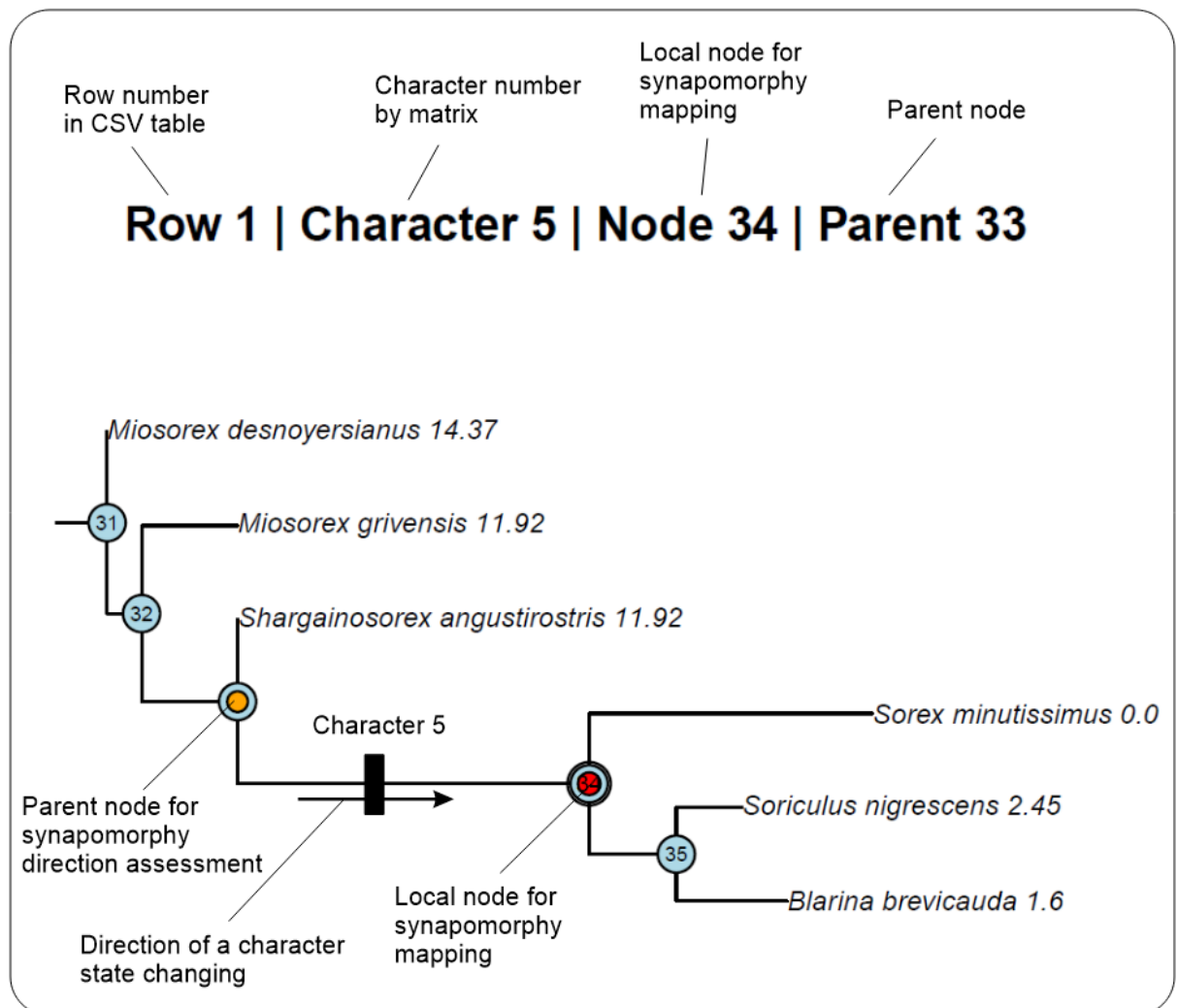


Supplementary Materials (S10)

for article

Rainer Hutterer's Naga shrew and the origin of modern soricids: a new
phylogenetic hypothesis

Diagram of the synapomorphy placement description



Supplementary materials: Table S10.1. Synapomorphy list obtained via stochastic mapping (Phytools; Mk-ER model) on the full MCC tree. Only state changes with high posterior probabilities (PP > 0.8) are considered; these probabilities refer to reconstructions carried out on the reduced taxon set (pruned trees). *Key:* **nT**, number of taxa after tree pruning in phytools.

Row	Ch. #	Node #	Parent Node	From	To	PP_node	Clade Size/nT	Character	Clade Tips
1	5	34	33	1	2	0.945	3/22	Mandibular condyle/Size: (0) small; (1) intermediate; (2) large	<i>Blarina, Soriculus, Sorex</i>
2	6	47	46	3	4	1.0	4/31	Mental foramen/Position: (0) far mesially from p4; (1) close mesially from p4; (2) below anterior root of p4; (3) below posterior root of p4; (4) below trigonid of m1; (5) distally trigonid of m1	<i>Blarina, Soriculus, Sorex, Shargainosorex</i>
3	6	39	38	2	4	0.91	2/31	<i>ibid.</i>	<i>Dobenflorinia, Miocrocidosorex</i>
4	8	21	20	2	3	0.825	5/14	Postsymphyseal foramen: (0) absent; (1) present mesially p4; (2) below p4; (3) below m1	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis</i>
5	9	28	27	0	1	0.87	3/18	Mandibular foramen: (0) below the middle part of the internal temporal fossa; (1) shifted at the level of the posterior corner of the fossa	<i>Blarina, Soriculus, Sorex</i>
6	12	38	37	2	3	0.985	3/24	Mandibular condyle/Shape: (0) undivided; (1) weakly divided; (2) divided with lingual fusion of the upper and lower facets [crocidurine-like]; (3) upper and lower facets clearly defined [soricine-like]	<i>Blarina, Soriculus, Sorex</i>
7	13	32	31	2	3	1.0	6/22	Mandibular condyle/Proportion: (0) elongated horizontally; (1) width slightly exceeds height; (2) height slightly	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus</i>

								exceeds width; (3) height significantly exceeds width	
8	19	34	33	0	1	1.0	17/26	i1 crown-root axis: (0) crown and root axes form an angle; (1) axes aligned	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus, Congosorex, Surdisorex, Nagasorex, M. okuensis, Crocidura, Suncus, M. meini, L. petersbuchense, L. dehmi, Florinia, Oligosorex</i>
9	20	36	35	0	1	0.98	19/28	i1 labial extension: (0) labial and lingual side are equal in length; (1) labial side is extended	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus, Congosorex, Surdisorex, Nagasorex, M. okuensis, Crocidura, Suncus, M. meini, Turiasorex, L. petersbuchense, L. dehmi, Florinia, Oligosorex, Clapasorex</i>
10	21	28	27	1	2	0.945	3/16	i1 root: (0) no grooves; (1) lingual groove only; (2) presence of lingual and labial grooves	<i>M. okuensis, Crocidura, Suncus</i>
11	22	44	43	3	4	1.0	4/29	Antemolars before p4: (0) more than four; (1) four; (2) three; (3) two; (4) one	<i>Blarina, Soriculus, Sorex, Shargainosorex</i>
12	22	42	41	2	3	1.0	6/29	<i>ibid.</i>	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus</i>
13	24	45	44	2	3	1.0	4/30	Morphology of p4: (0) molarized; (1) only labial crest; (2) two sub-equal crests; (3) labial crest longer than lingual crest	<i>Blarina, Soriculus, Sorex, Shargainosorex</i>
14	24	55	51	2	3	1.0	2/30	<i>ibid.</i>	<i>Crocidura, Suncus</i>
15	24	46	45	3	1	0.97	3/30	<i>ibid.</i>	<i>Blarina, Soriculus, Sorex</i>
16	25	31	30	0	1	0.98	24/28	p4 crown symmetry in occlusal view: (0) lingual and labial sides appear symmetrical; (1) buccal part extends posteriorly	<i>Aralosorex, Dobenflorinia, Carposorex, Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus, Congosorex, Surdisorex, Nagasorex, M. okuensis, Crocidura, Suncus, M. meini, Turiasorex, L. petersbuchense, L. dehmi, Florinia, Oligosorex, Clapasorex, Crocidosorex, Meingensorex</i>
17	26	43	42	0	1	0.985	3/28	p4 posterior sulcus: (0) present; (1) absence	<i>Blarina, Soriculus, Sorex</i>
18	29	44	40	0	1	0.93	21/33	m1-m2/protoconid: (0) distolabial fold reaches the ectocingulid [low	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus, Congosorex,</i>

								hypoflexid]; (1) the fold does not reach the cingulid [high hypoflexid]	<i>Surdisorex, Nagasorex, M. okuensis, Crocidura, Suncus, M. meini, Turiasorex, L. petersbuchense, L. dehmi, Florinia, Oligosorex, Clapasorex, Crocidosorex, Meingensorex</i>
19	30	49	48	0	1	1.0	4/32	m1-m2/labial notch: (0) present; (1) absent	<i>Blarina, Soriculus, Sorex, Shargainosorex</i>
20	30	50	49	1	0	0.915	3/32	<i>ibid.</i>	<i>Blarina, Soriculus, Sorex</i>
21	33	49	48	1	2	1.0	6/33	m1-m2/postentoconid valley: (0) hypolophid connects entoconid [valley absent]; (1) free-ended hypolophid with narrow valley; (2) free-ended hypolophid with wide valley	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus</i>
22	35	49	48	0	1	1.0	6/33	m1-m2/entoconid: (0) conical; (1) crested	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus</i>
23	35	65	36	0	1	0.96	2/33	<i>ibid.</i>	<i>Domnina, Heterosorex</i>
24	35	57	56	0	1	0.9	6/33	<i>ibid.</i>	<i>Congosorex, Surdisorex, Nagasorex, M. okuensis, Crocidura, Suncus</i>
25	36	51	50	0	1	1.0	4/33	m1-m2/hypoconulid: (0) absent; (1) present	<i>Blarina, Soriculus, Sorex, Shargainosorex</i>
26	45	41	40	0	1	0.825	3/26	P4/parastylar crest; (0) absent; (1) present	<i>Blarina, Soriculus, Sorex</i>
27	52	49	48	0	1	1.0	4/28	M1/mesostyle condition: (0) confluent; (1) divided	<i>Congosorex, Surdisorex, Nagasorex, M. okuensis</i>
28	55	40	39	0	1	1.0	6/28	M1/hypocone: (0) present; (1) absent, incorporated into the hypoconal flange ridge	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus</i>
29	57	53	47	0	1	1.0	2/28	M1/postcingulum condition: (0) continuous through the posterior emargination; (1) cingulum with gap	<i>M. meini, Turiasorex</i>
30	59	41	40	0	1	1.0	6/28	M2/hypocone: (0) present; (1) absent, incorporated into the hypoconal flange ridge	<i>Blarina, Soriculus, Sorex, Shargainosorex, M. grivensis, M. desnoyersianus</i>

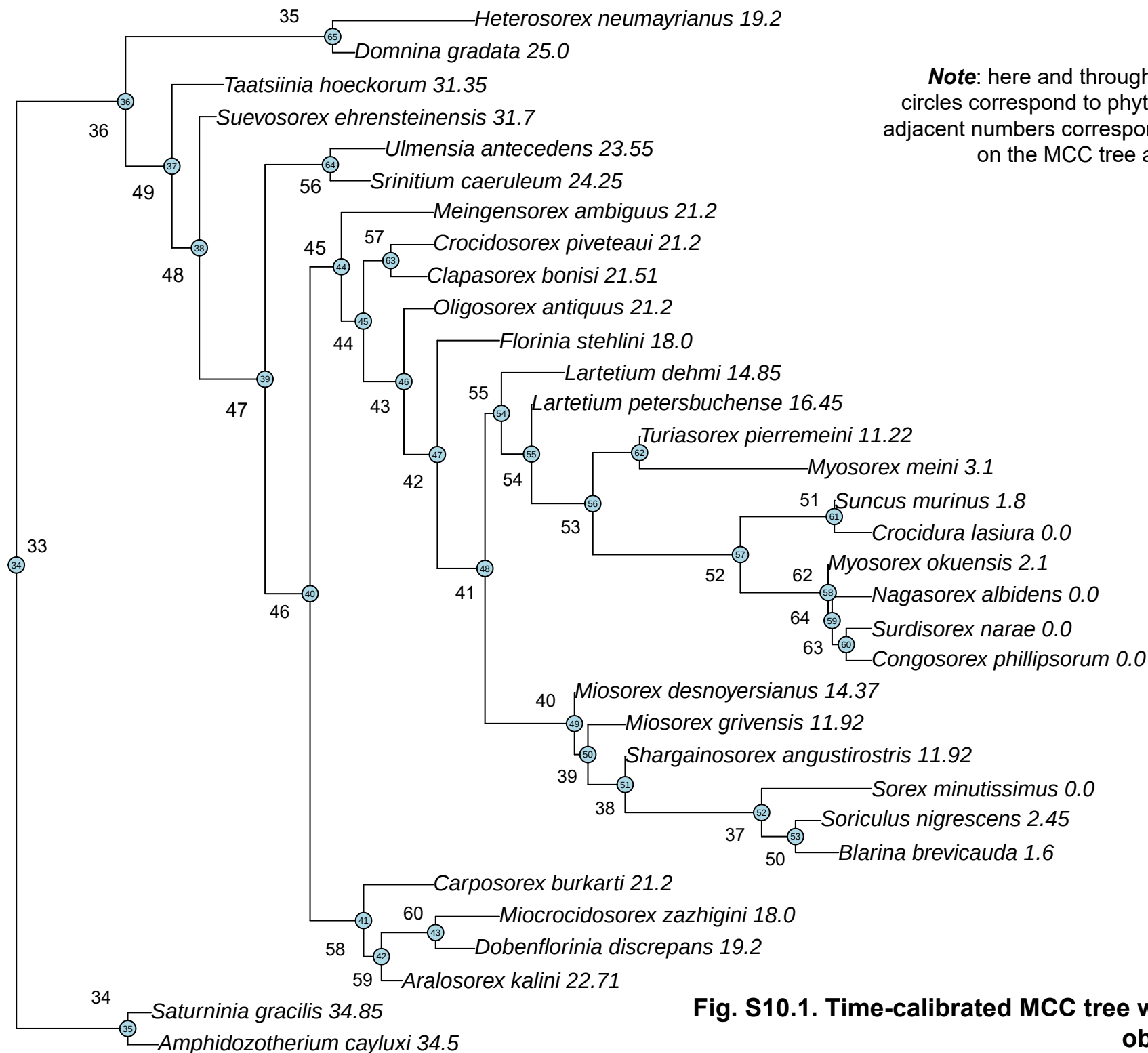
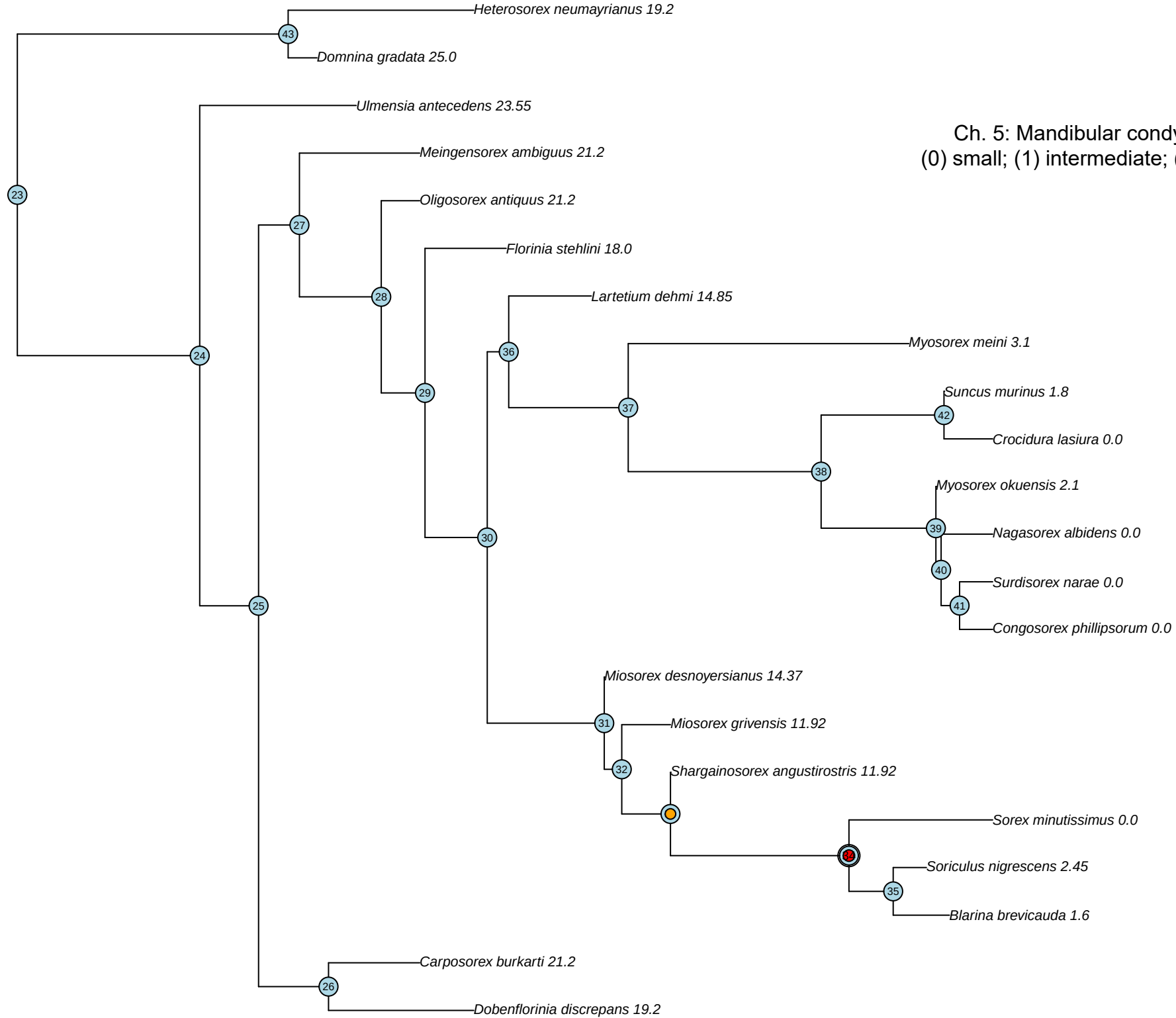
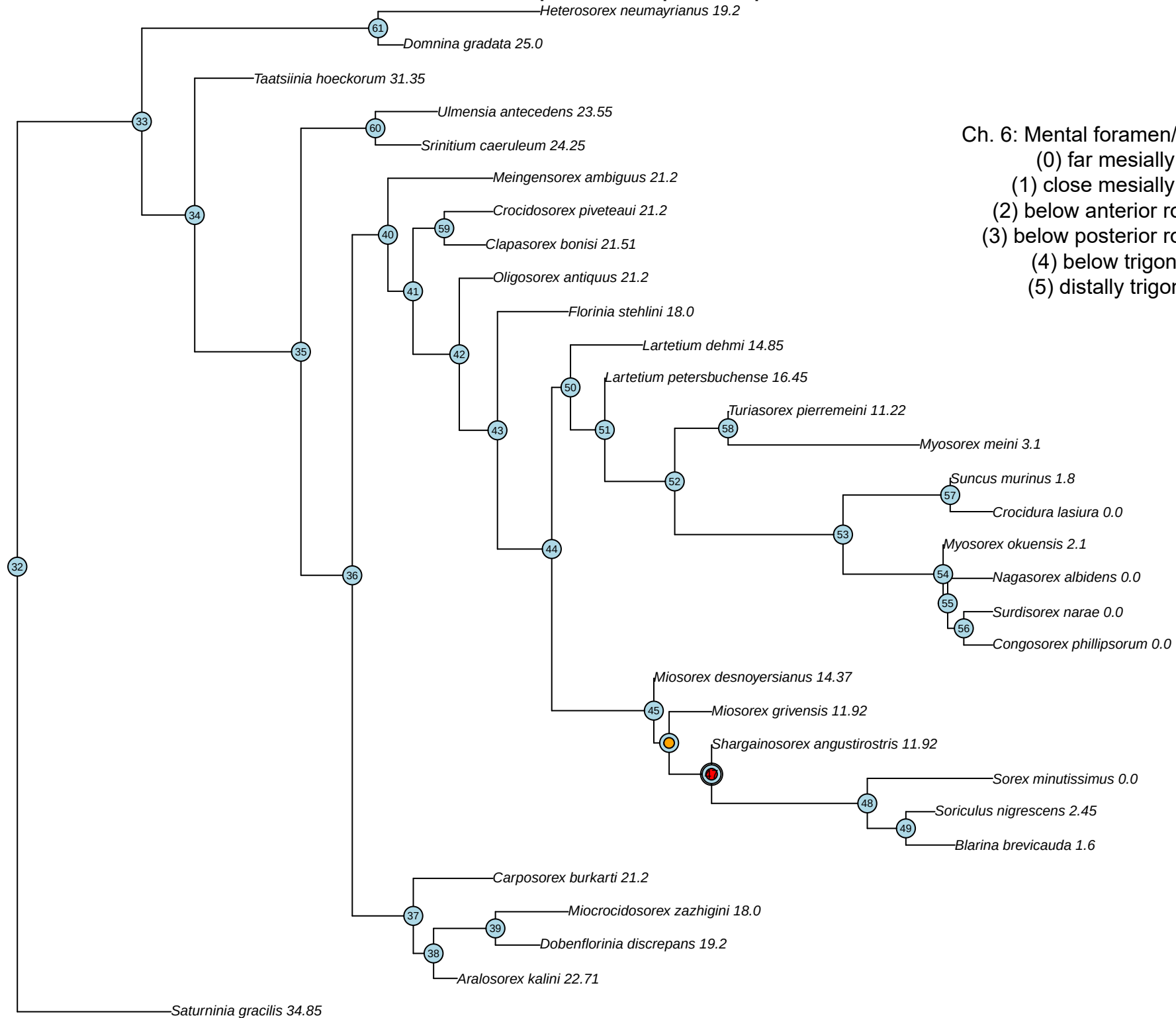


Fig. S10.1. Time-calibrated MCC tree with node labels as obtained in phytools

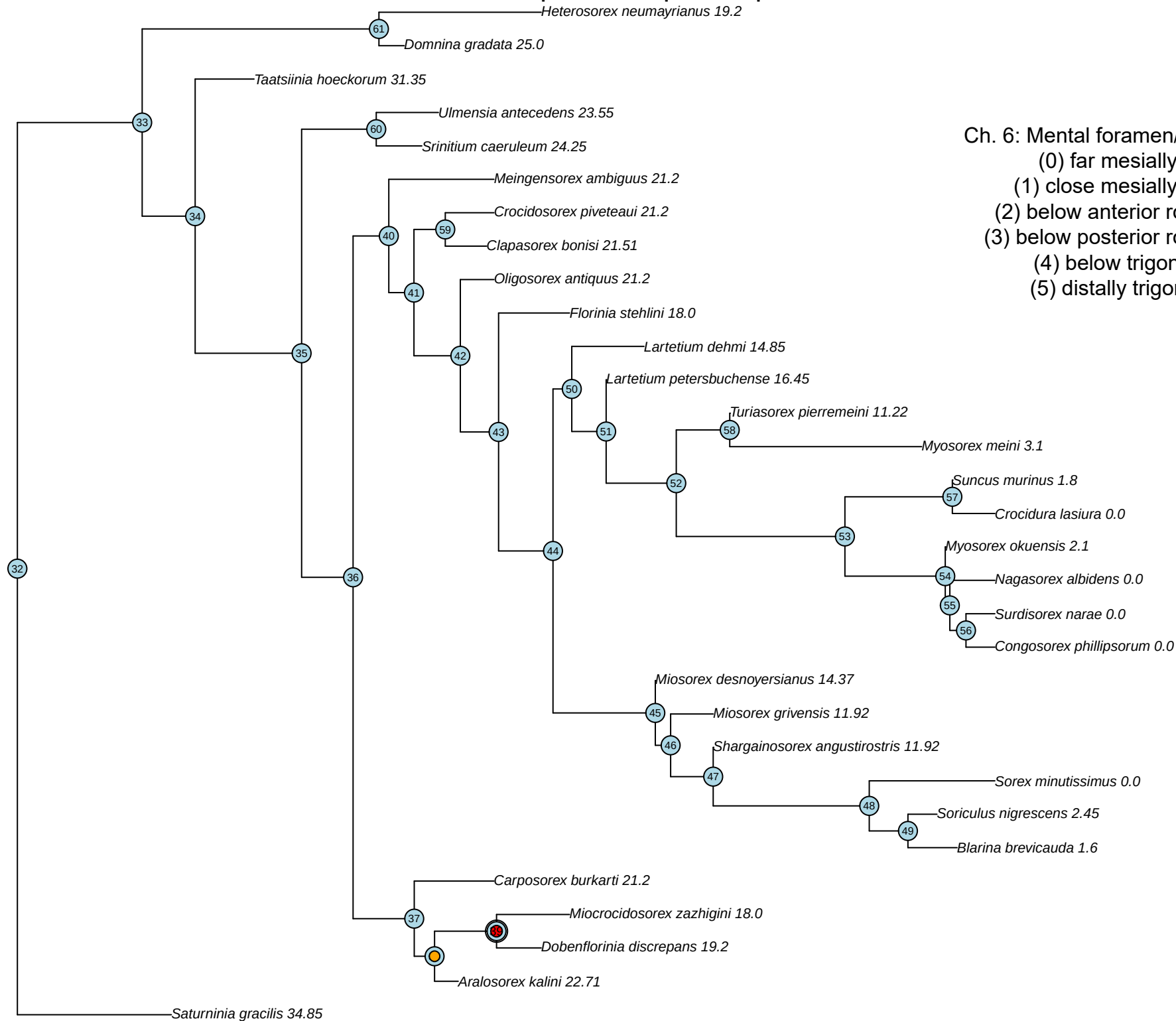
Row 1 | Character 5 | Node 34 | Parent 33



Row 2 | Character 6 | Node 47 | Parent 46

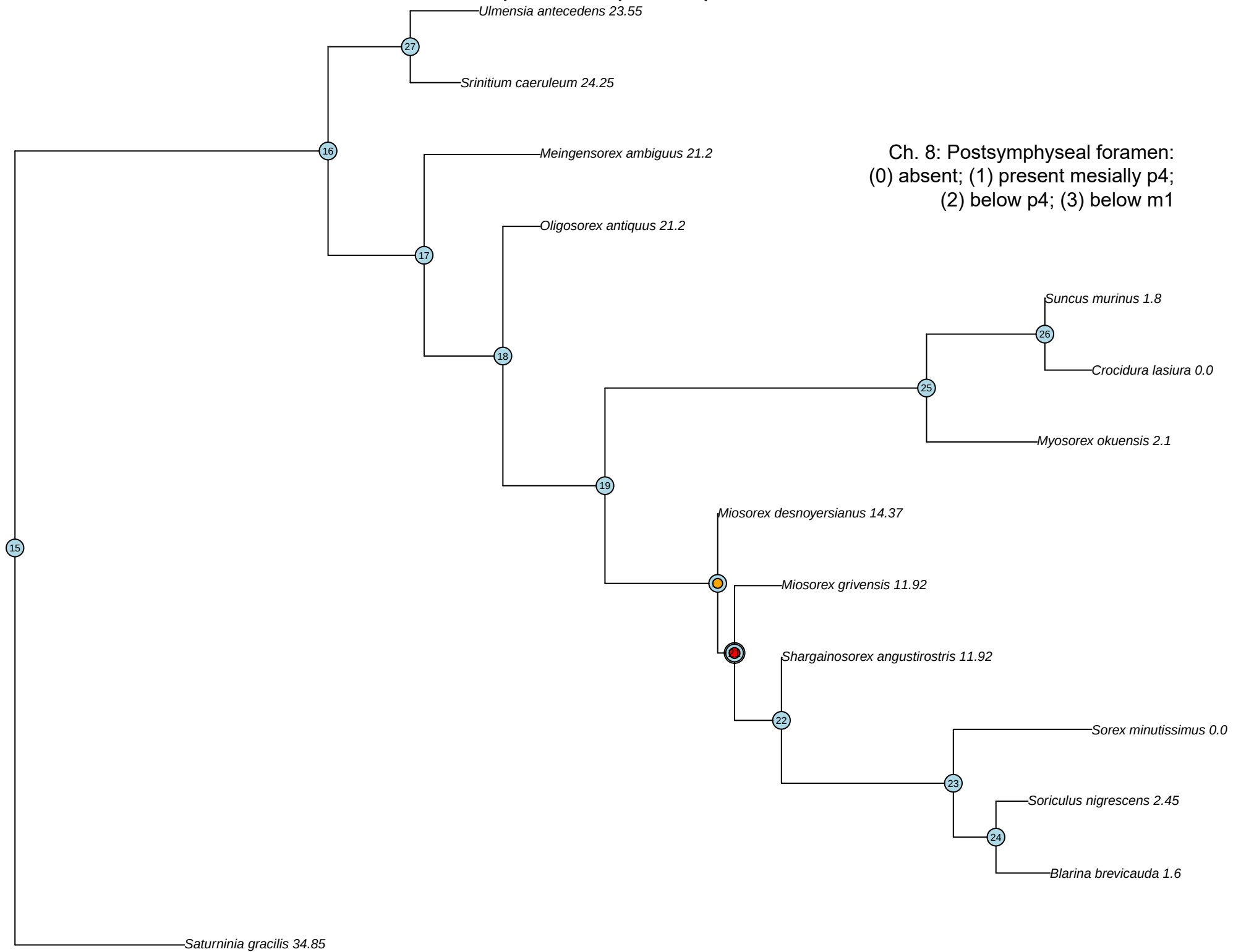


Row 3 | Character 6 | Node 39 | Parent 38

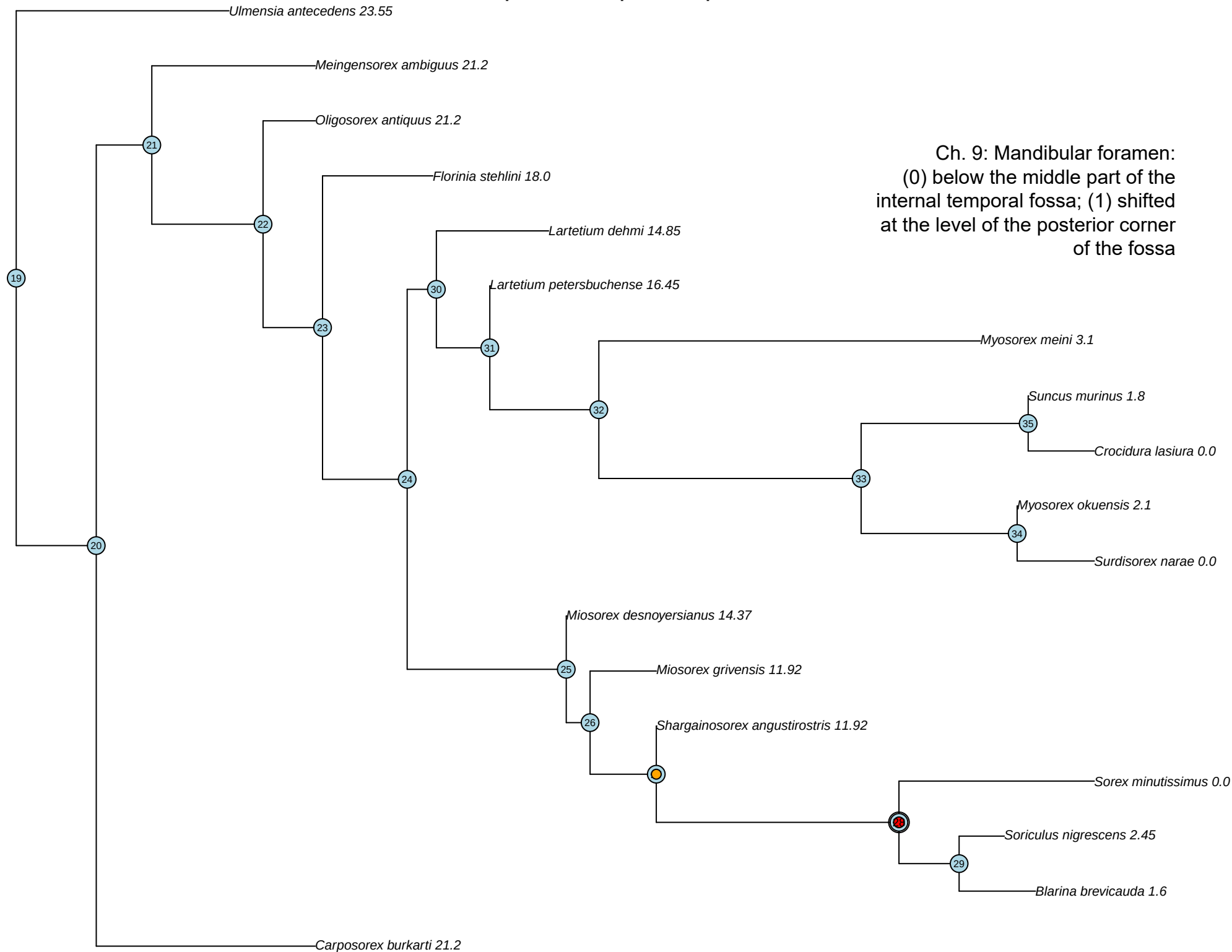


Ch. 6: Mental foramen/Position:
 (0) far mesially from p4;
 (1) close mesially from p4;
 (2) below anterior root of p4;
 (3) below posterior root of p4;
 (4) below trigonid of m1;
 (5) distally trigonid of m1

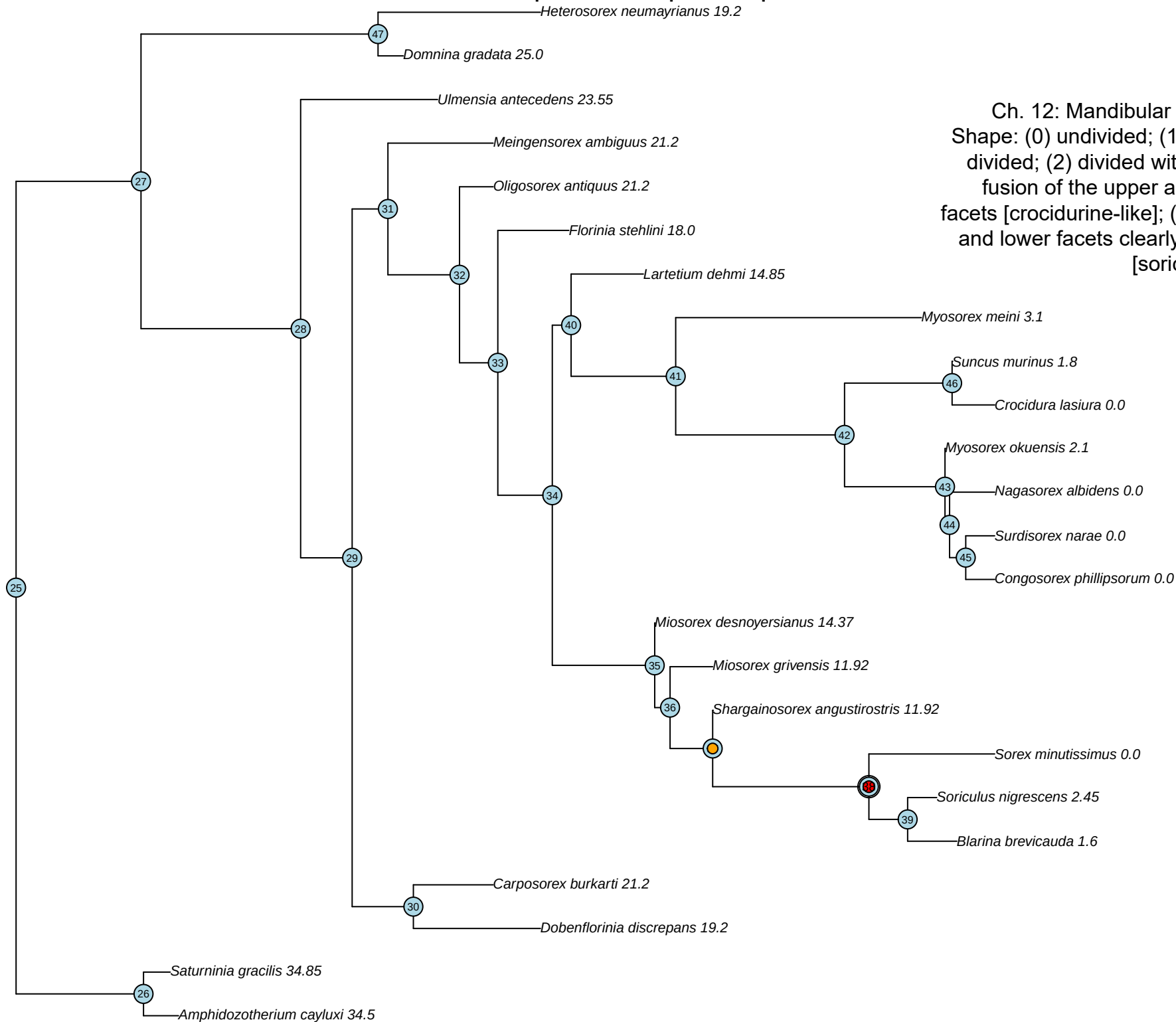
Row 4 | Character 8 | Node 21 | Parent 20



Row 5 | Character 9 | Node 28 | Parent 27

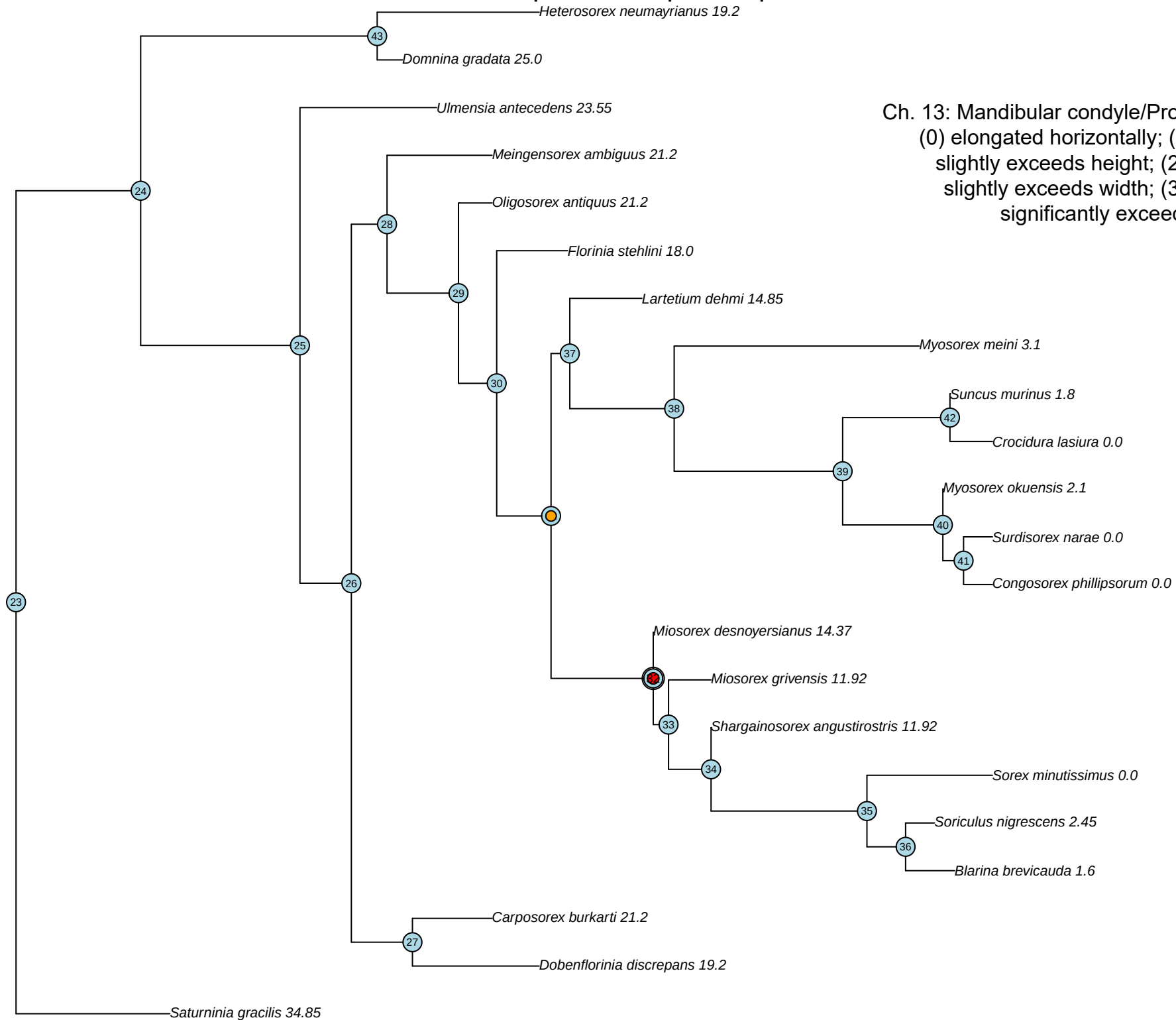


Row 6 | Character 12 | Node 38 | Parent 37



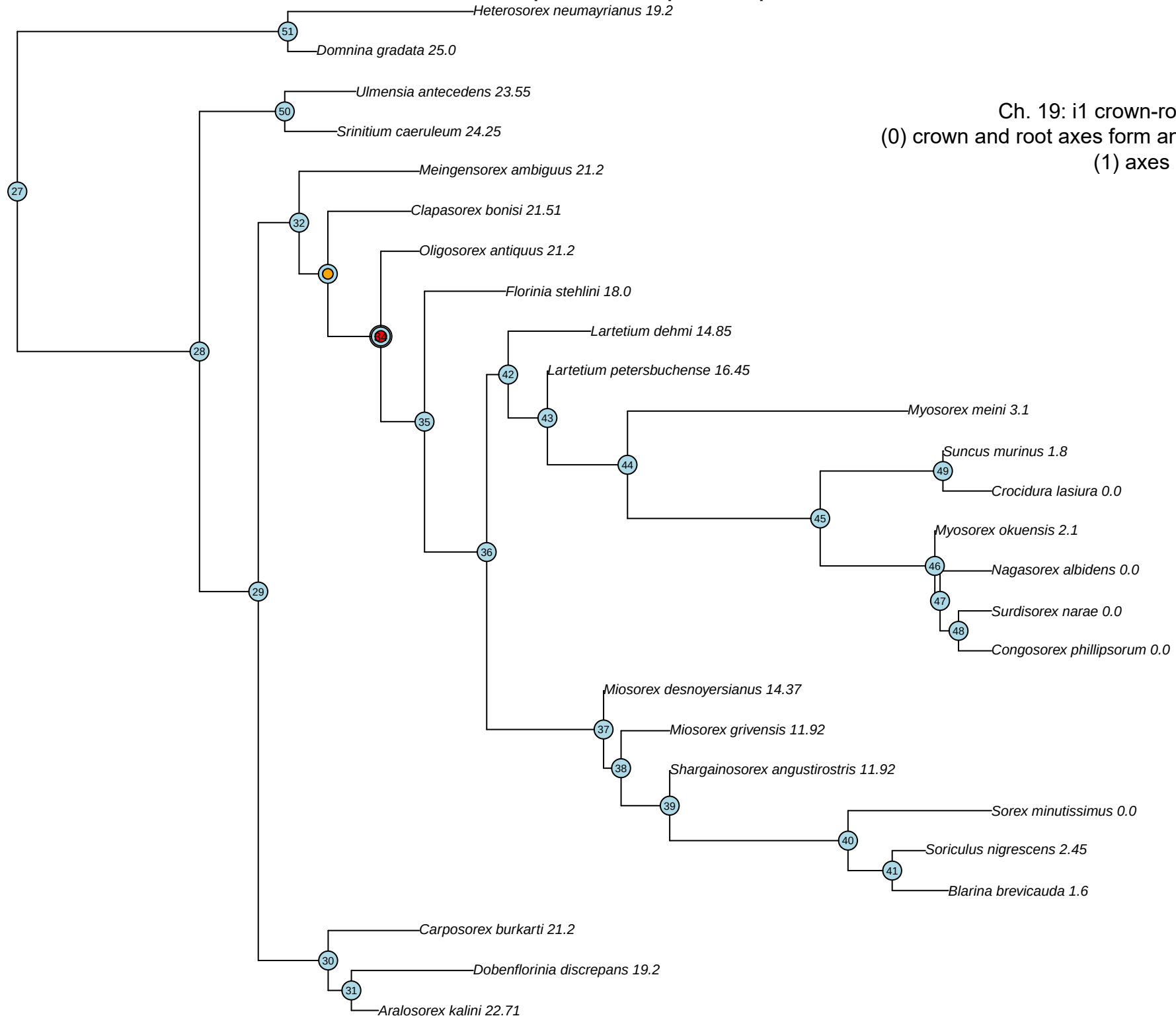
Ch. 12: Mandibular condyle/
Shape: (0) undivided; (1) weakly
divided; (2) divided with lingual
fusion of the upper and lower
facets [crocidurine-like]; (3) upper
and lower facets clearly defined
[soricine-like]

Row 7 | Character 13 | Node 32 | Parent 31



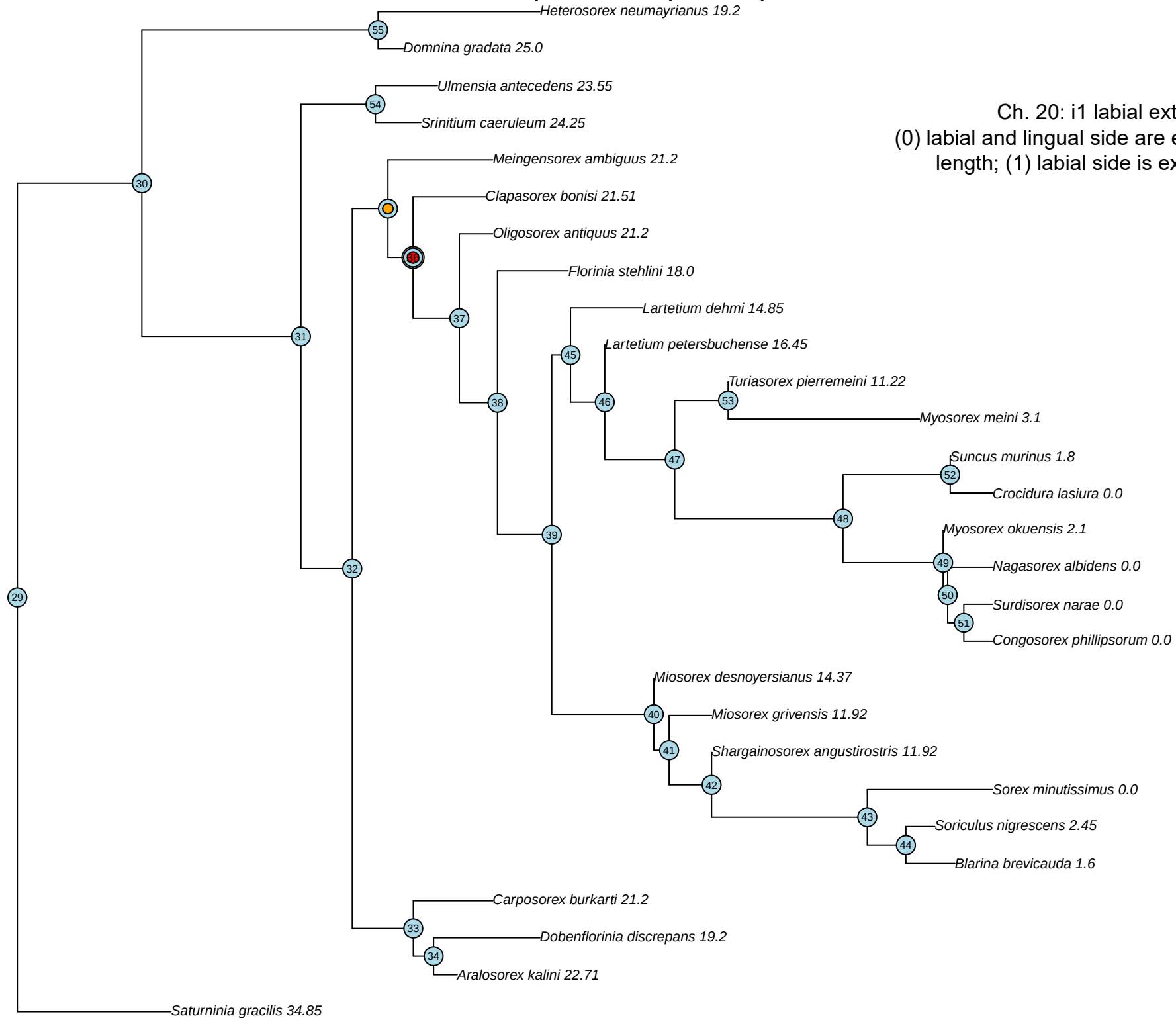
Ch. 13: Mandibular condyle/Proportion:
 (0) elongated horizontally; (1) width
 slightly exceeds height; (2) height
 slightly exceeds width; (3) height
 significantly exceeds width

Row 8 | Character 19 | Node 34 | Parent 33



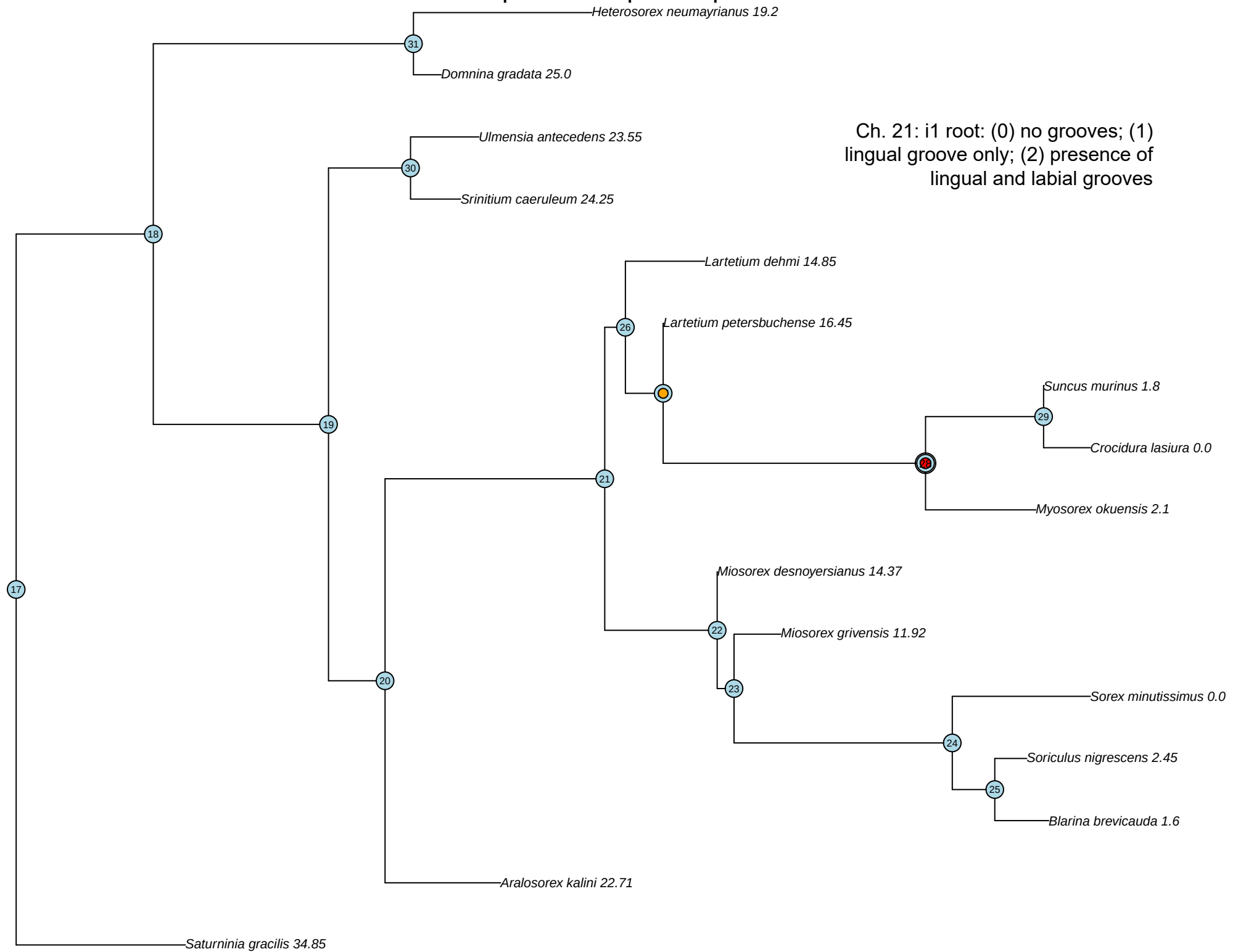
Ch. 19: i1 crown-root axis:
 (0) crown and root axes form an angle;
 (1) axes aligned

Row 9 | Character 20 | Node 36 | Parent 35



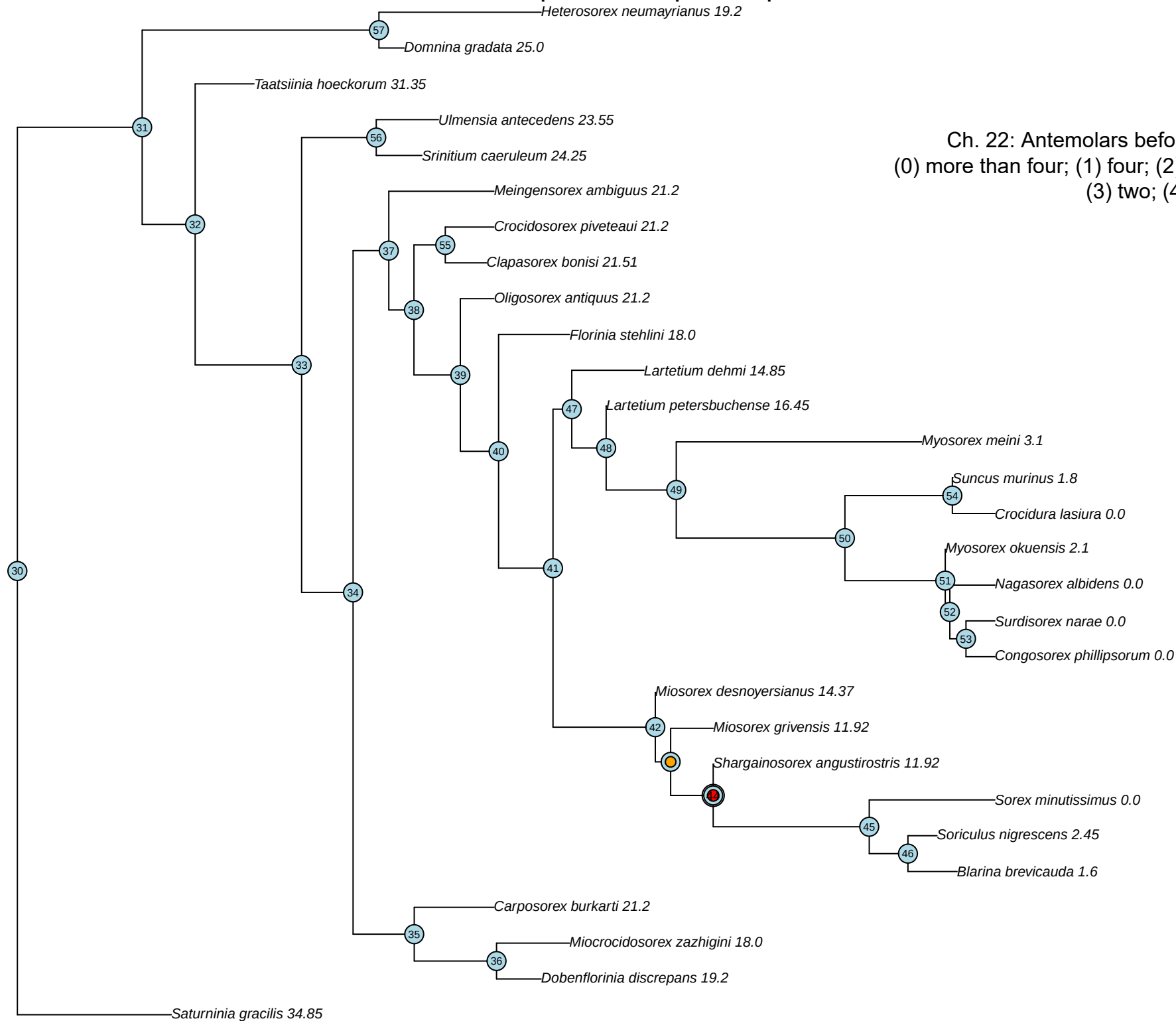
Ch. 20: i1 labial extension:
(0) labial and lingual side are equal in length; (1) labial side is extended

Row 10 | Character 21 | Node 28 | Parent 27



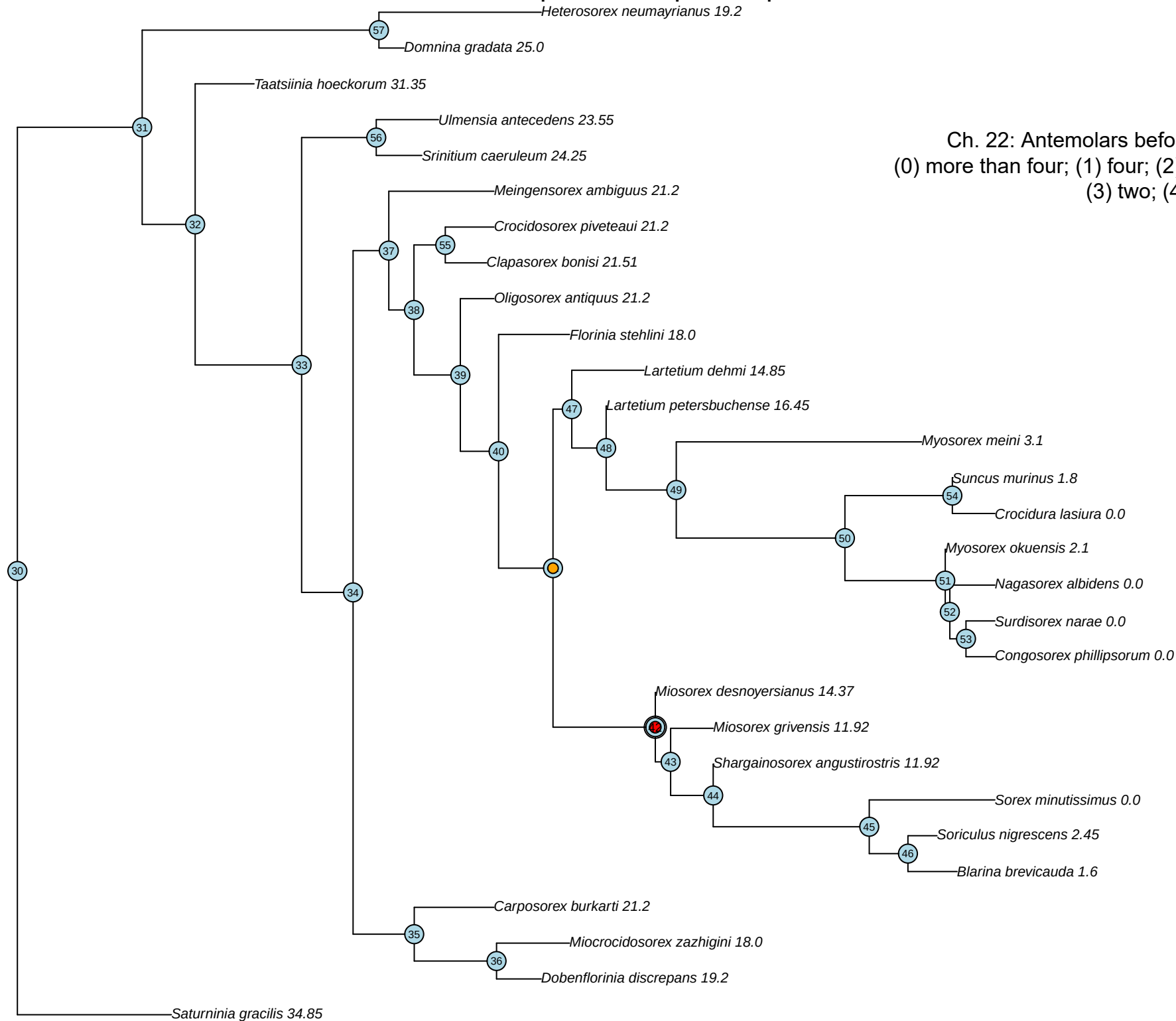
Ch. 21: i1 root: (0) no grooves; (1) lingual groove only; (2) presence of lingual and labial grooves

Row 11 | Character 22 | Node 44 | Parent 43



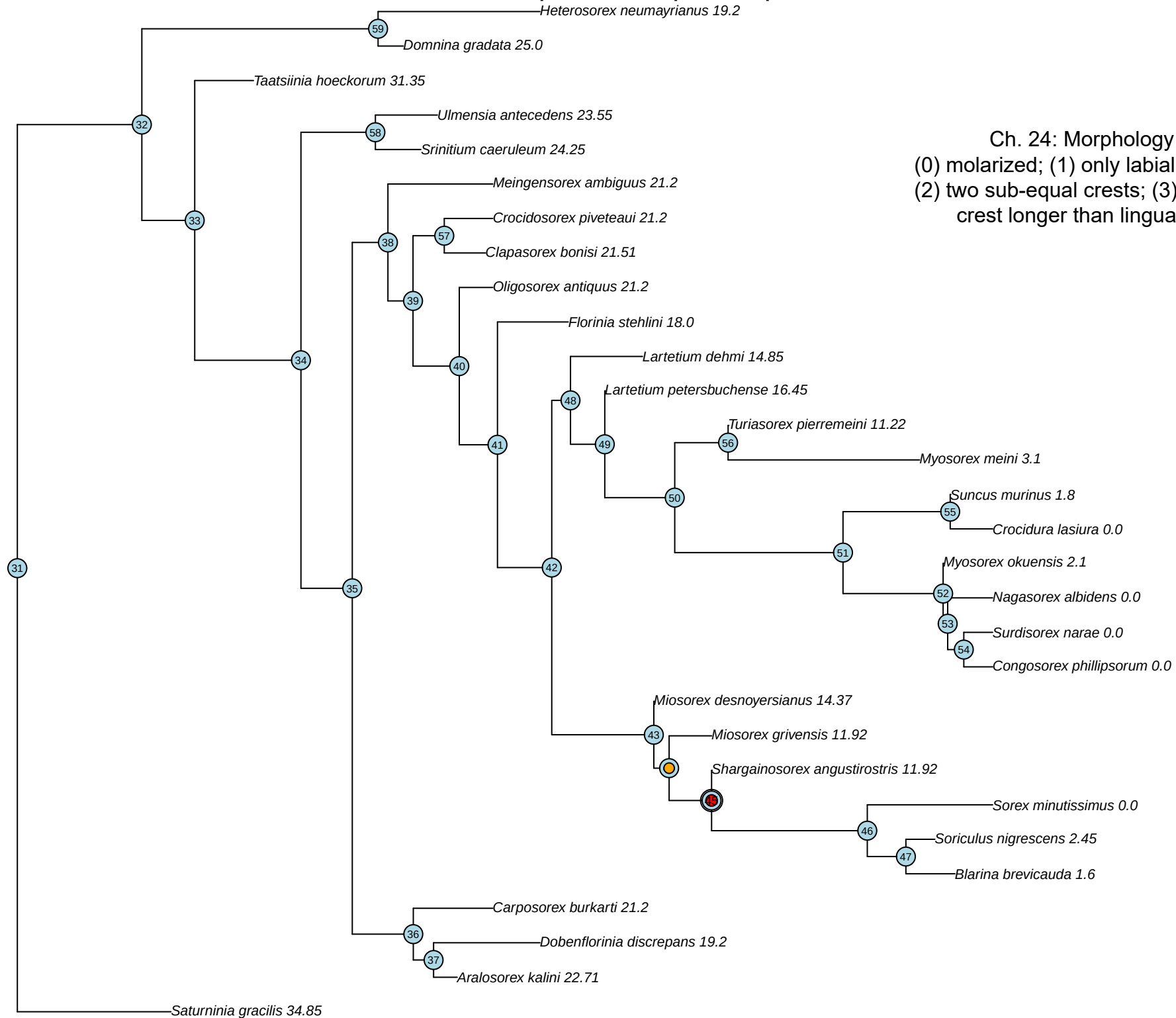
Ch. 22: Antemolars before p4:
 (0) more than four; (1) four; (2) three;
 (3) two; (4) one

Row 12 | Character 22 | Node 42 | Parent 41



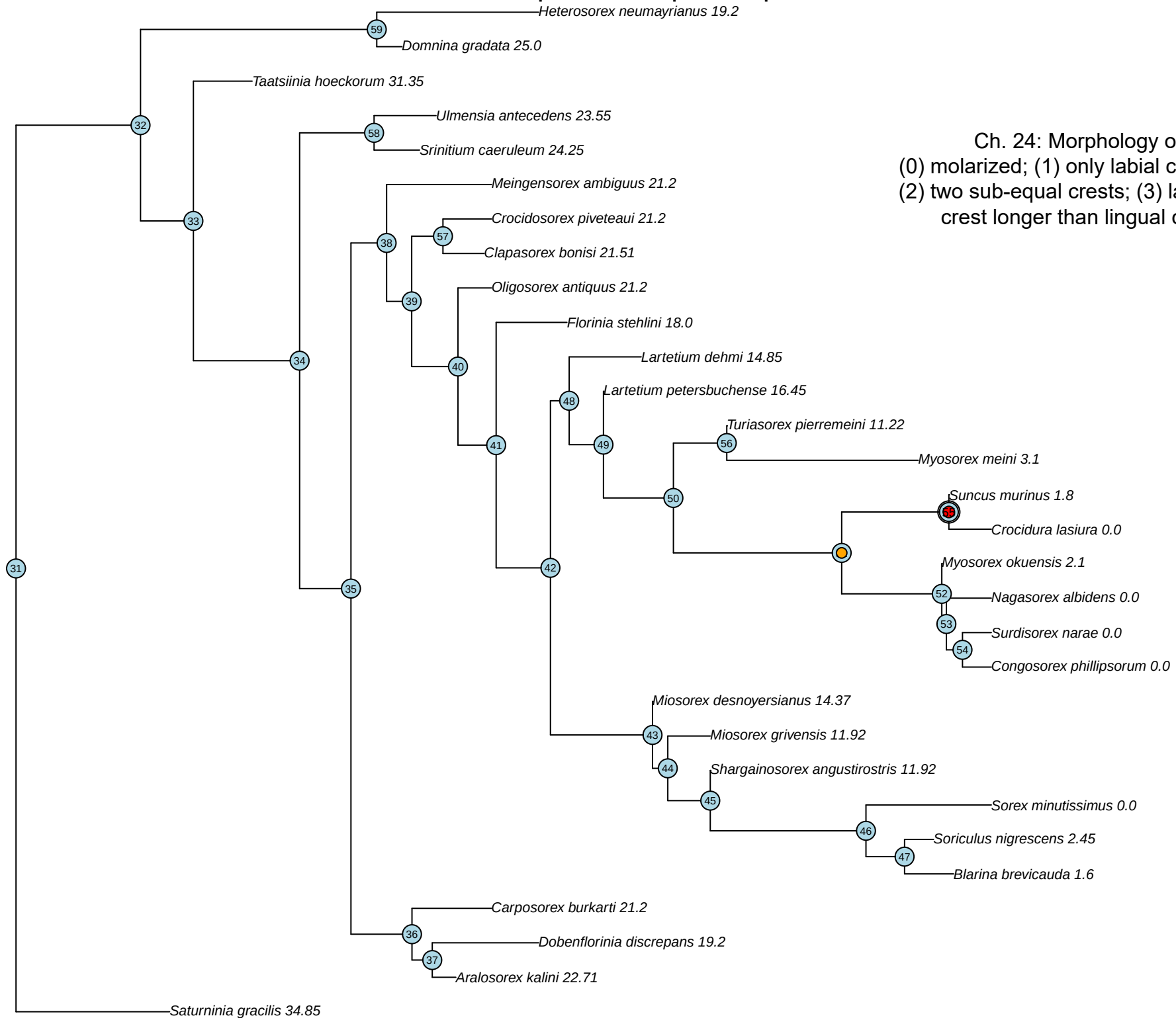
Ch. 22: Antemolars before p4:
 (0) more than four; (1) four; (2) three;
 (3) two; (4) one

Row 13 | Character 24 | Node 45 | Parent 44



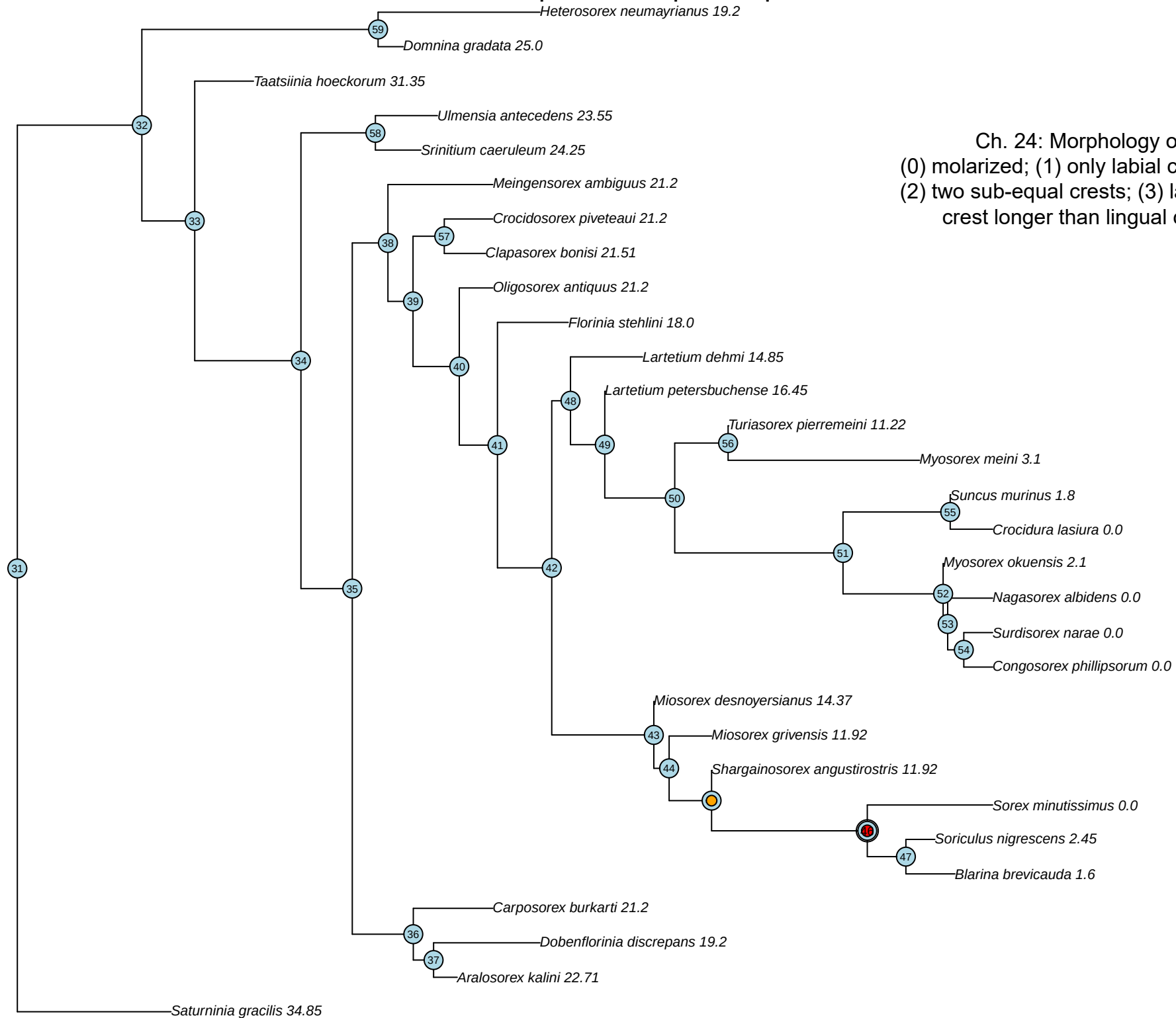
Ch. 24: Morphology of p4:
 (0) molarized; (1) only labial crest;
 (2) two sub-equal crests; (3) labial
 crest longer than lingual crest

Row 14 | Character 24 | Node 55 | Parent 51



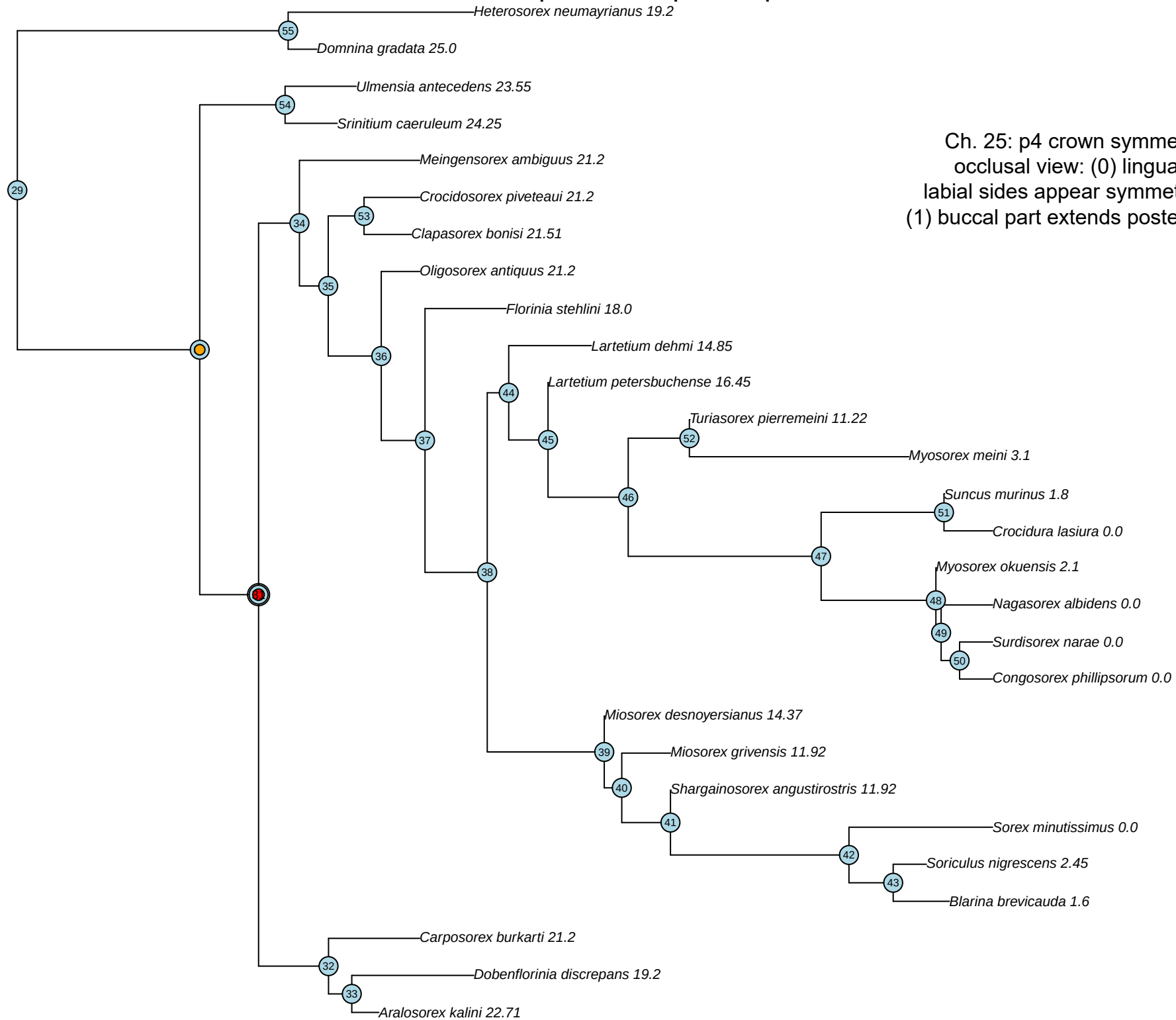
Ch. 24: Morphology of p4:
 (0) molarized; (1) only labial crest;
 (2) two sub-equal crests; (3) labial
 crest longer than lingual crest

Row 15 | Character 24 | Node 46 | Parent 45



Ch. 24: Morphology of p4:
 (0) molarized; (1) only labial crest;
 (2) two sub-equal crests; (3) labial
 crest longer than lingual crest

Row 16 | Character 25 | Node 31 | Parent 30



Ch. 25: p4 crown symmetry in occlusal view: (0) lingual and labial sides appear symmetrical; (1) buccal part extends posteriorly

Row 17 | Character 26 | Node 43 | Parent 42

Heterosorex neumayrianus 19.2

Domnina gradata 25.0

Taatsiinina hoeckorum 31.35

Ulmensia antecessens 23.55

Meingensorex ambiguus 21.2

Crocidosorex piveteaui 21.2

Clapasorex bonisi 21.51

Oligosorex antiquus 21.2

Florinia stehlini 18.0

Lartetium dehmi 14.85

Lartetium petersbuchense 16.45

Turiasorex pierremeini 11.22

Myosorex meini 3.1

Suncus murinus 1.8

Crocidura lasiura 0.0

Myosorex okuensis 2.1

Nagasurex albidens 0.0

Surdisorex narae 0.0

Congosorex phillipsorum 0.0

Miosorex desnoyersianus 14.37

Miosorex grivensis 11.92

Shargainosorex angustirostris 11.92

Sorex minutissimus 0.0

Soriculus nigrescens 2.45

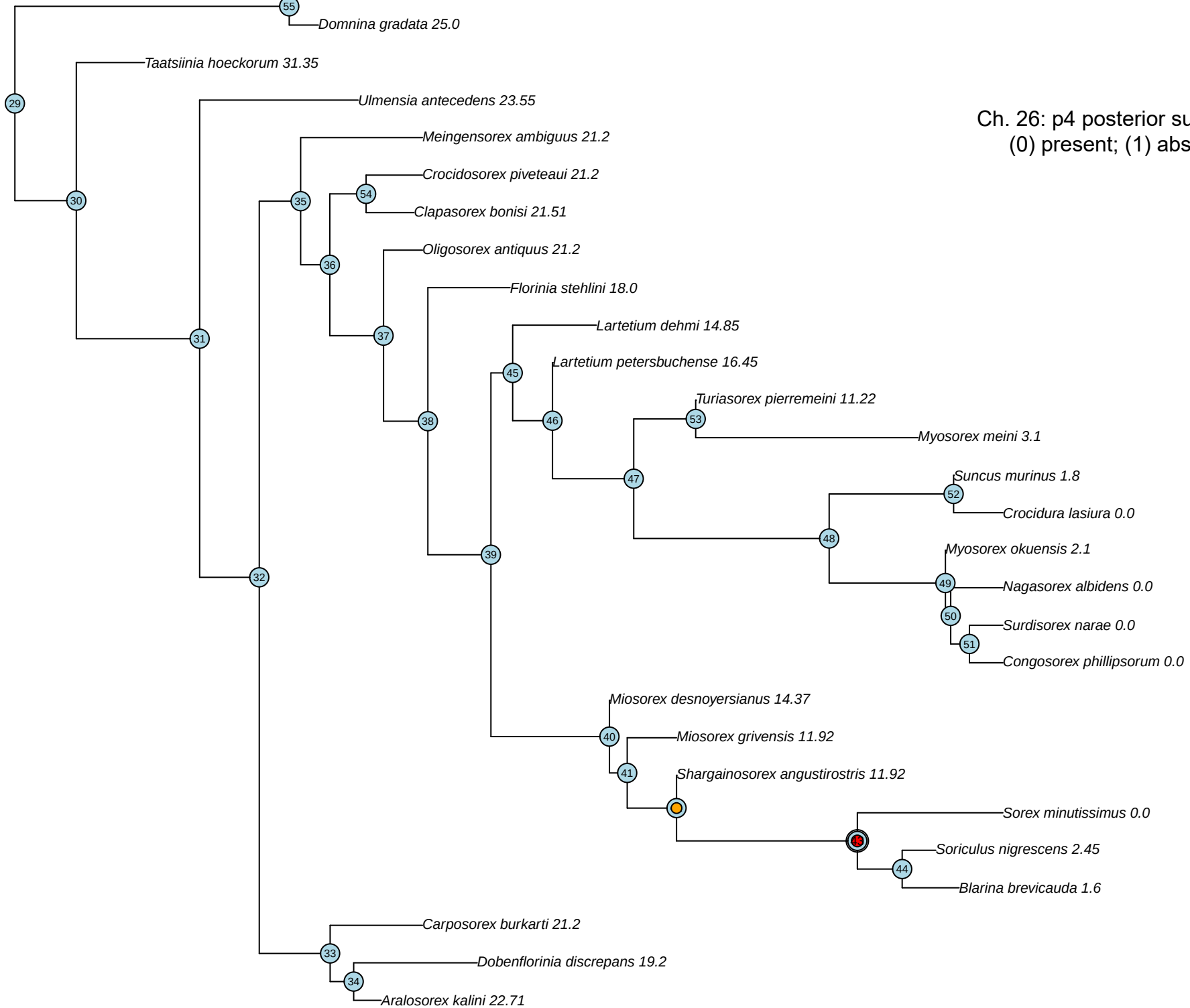
Blarina brevicauda 1.6

Carposorex burkarti 21.2

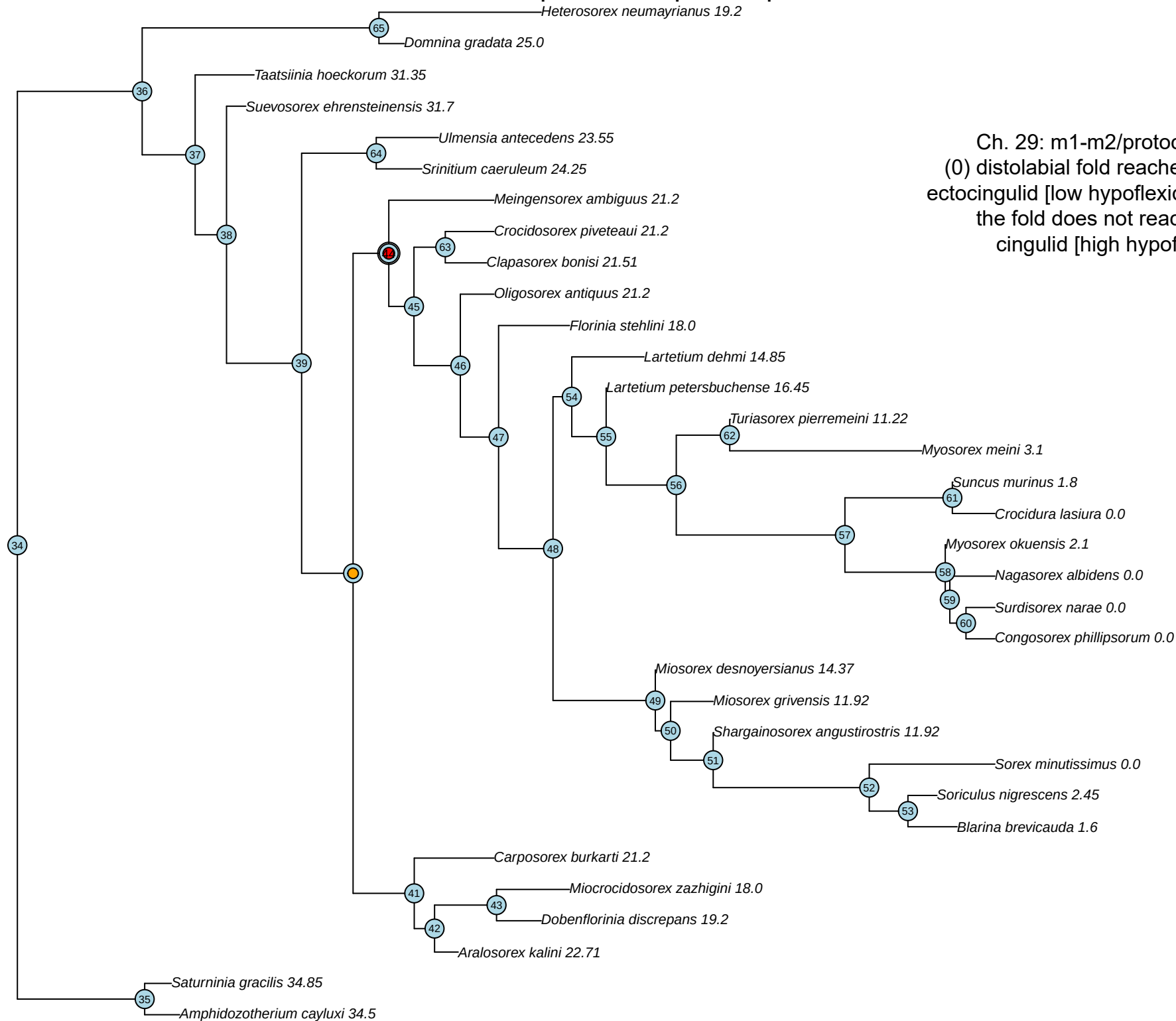
Dobenflorinia discrepans 19.2

Aralosorex kalini 22.71

Ch. 26: p4 posterior sulcus:
(0) present; (1) absence

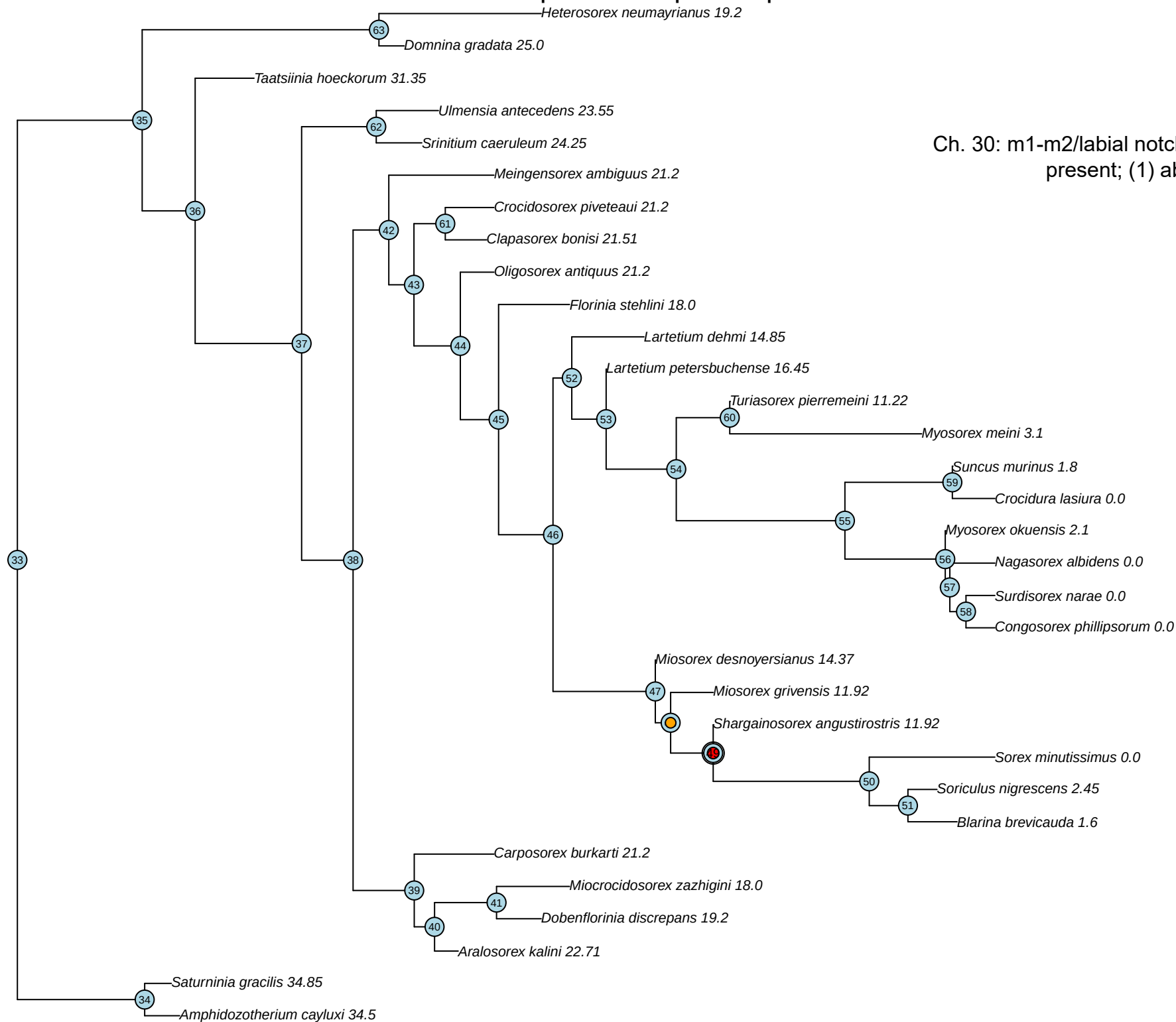


Row 18 | Character 29 | Node 44 | Parent 40



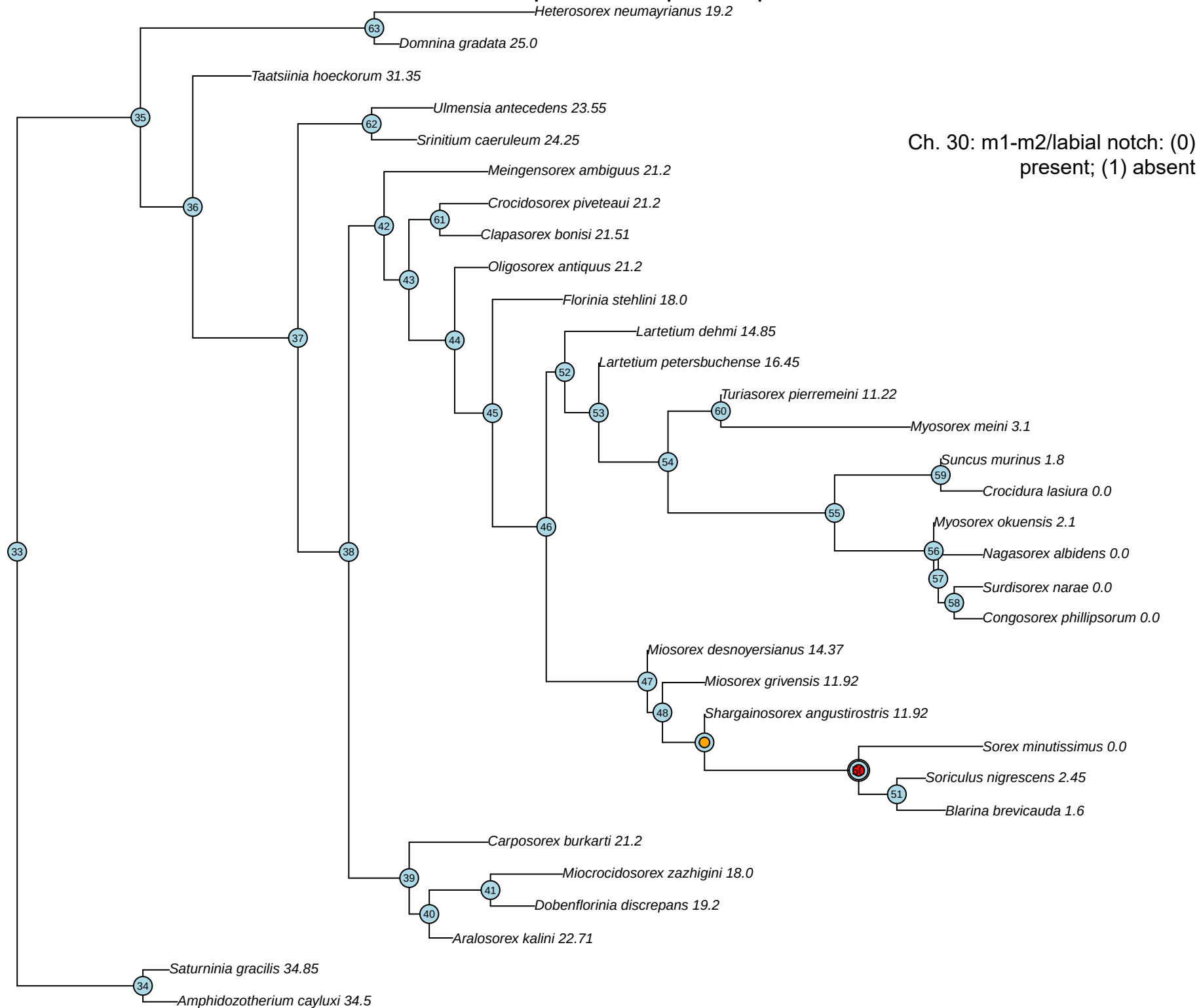
Ch. 29: m1-m2/protoconid:
 (0) distolabial fold reaches the
 ectocingulid [low hypoflexid]; (1)
 the fold does not reach the
 cingulid [high hypoflexid]

Row 19 | Character 30 | Node 49 | Parent 48

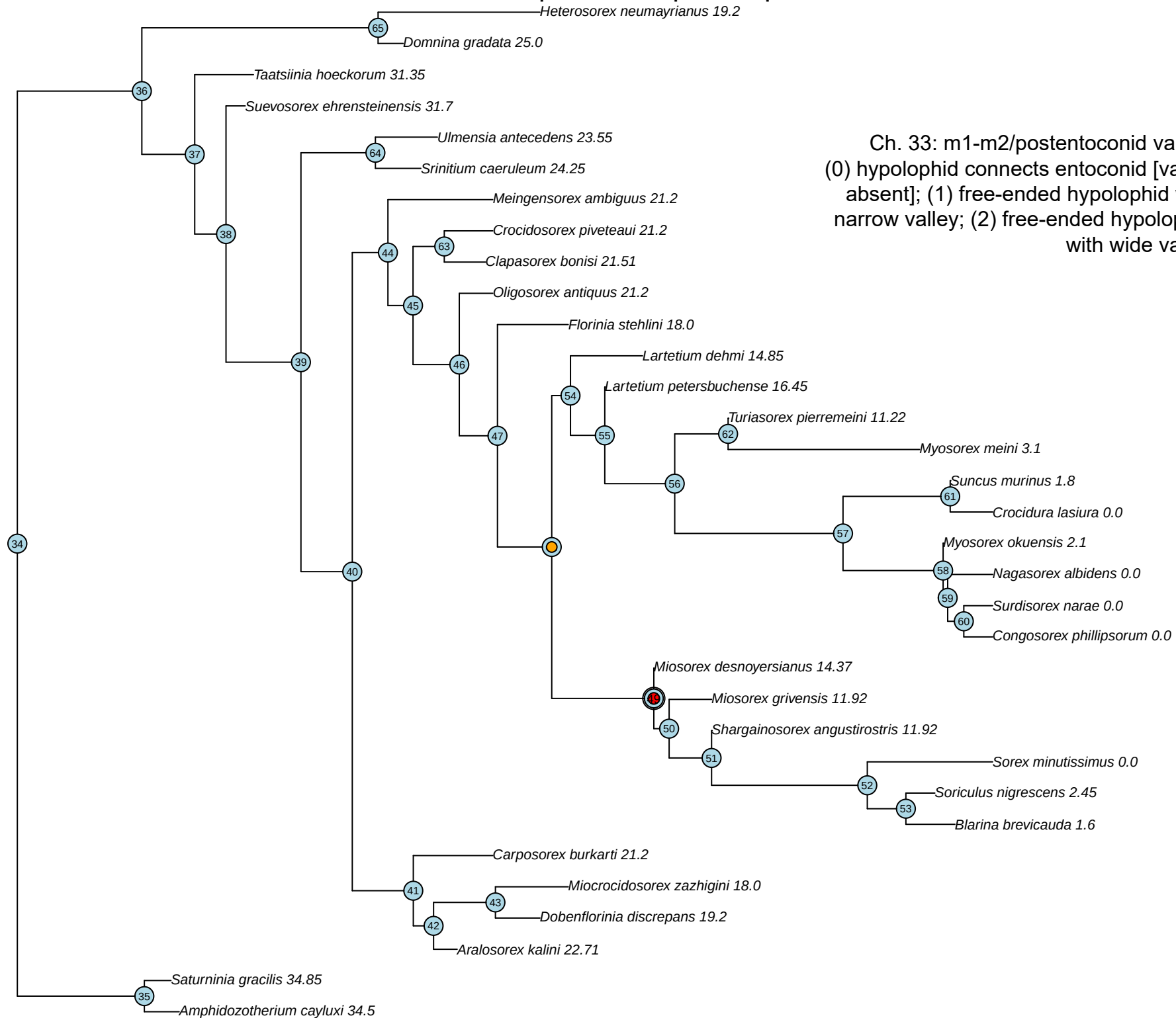


Ch. 30: m1-m2/labial notch: (0) present; (1) absent

Row 20 | Character 30 | Node 50 | Parent 49

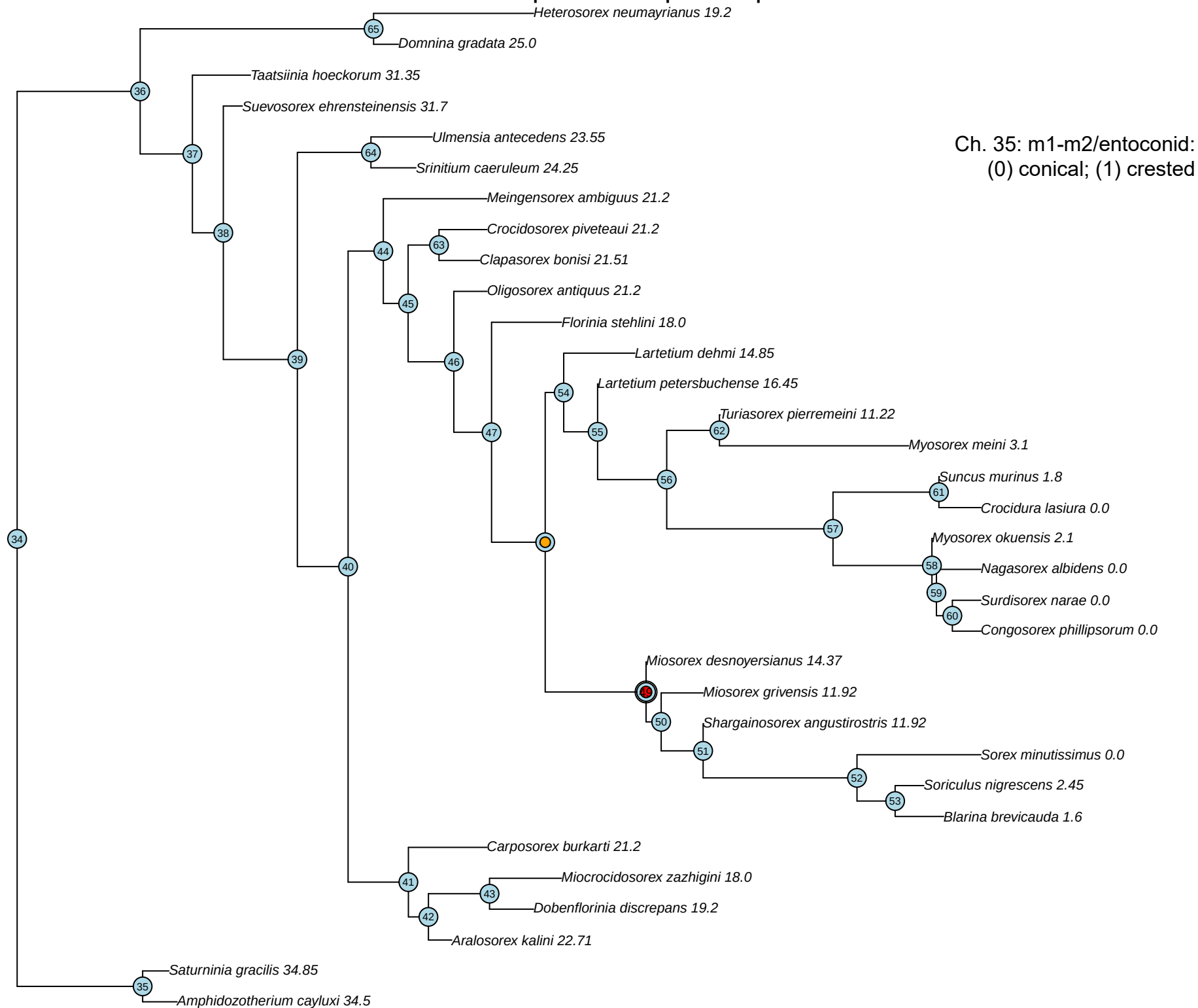


Row 21 | Character 33 | Node 49 | Parent 48

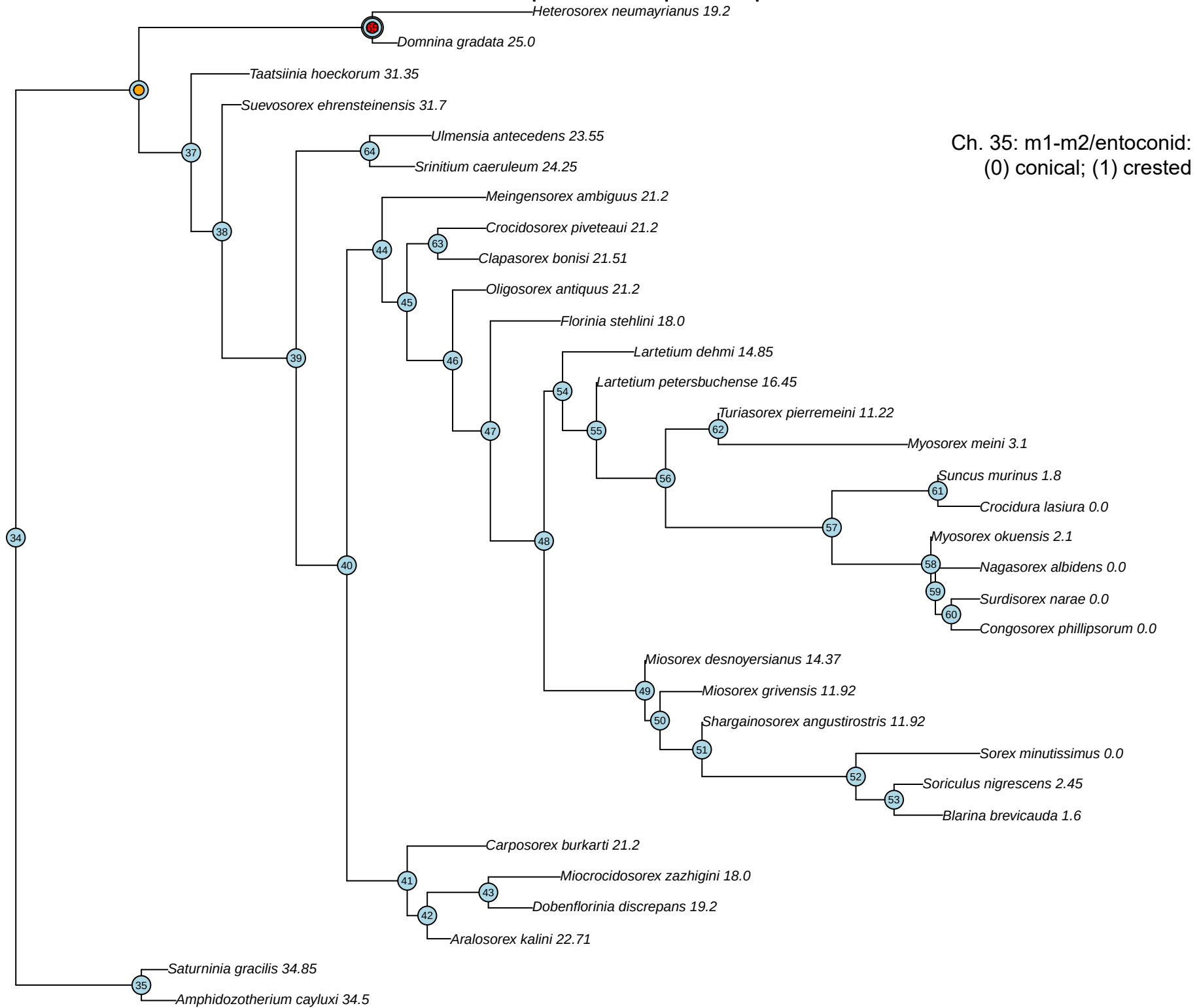


Ch. 33: m1-m2/postentoconid valley:
 (0) hypolophid connects entoconid [valley
 absent]; (1) free-ended hypolophid with
 narrow valley; (2) free-ended hypolophid
 with wide valley

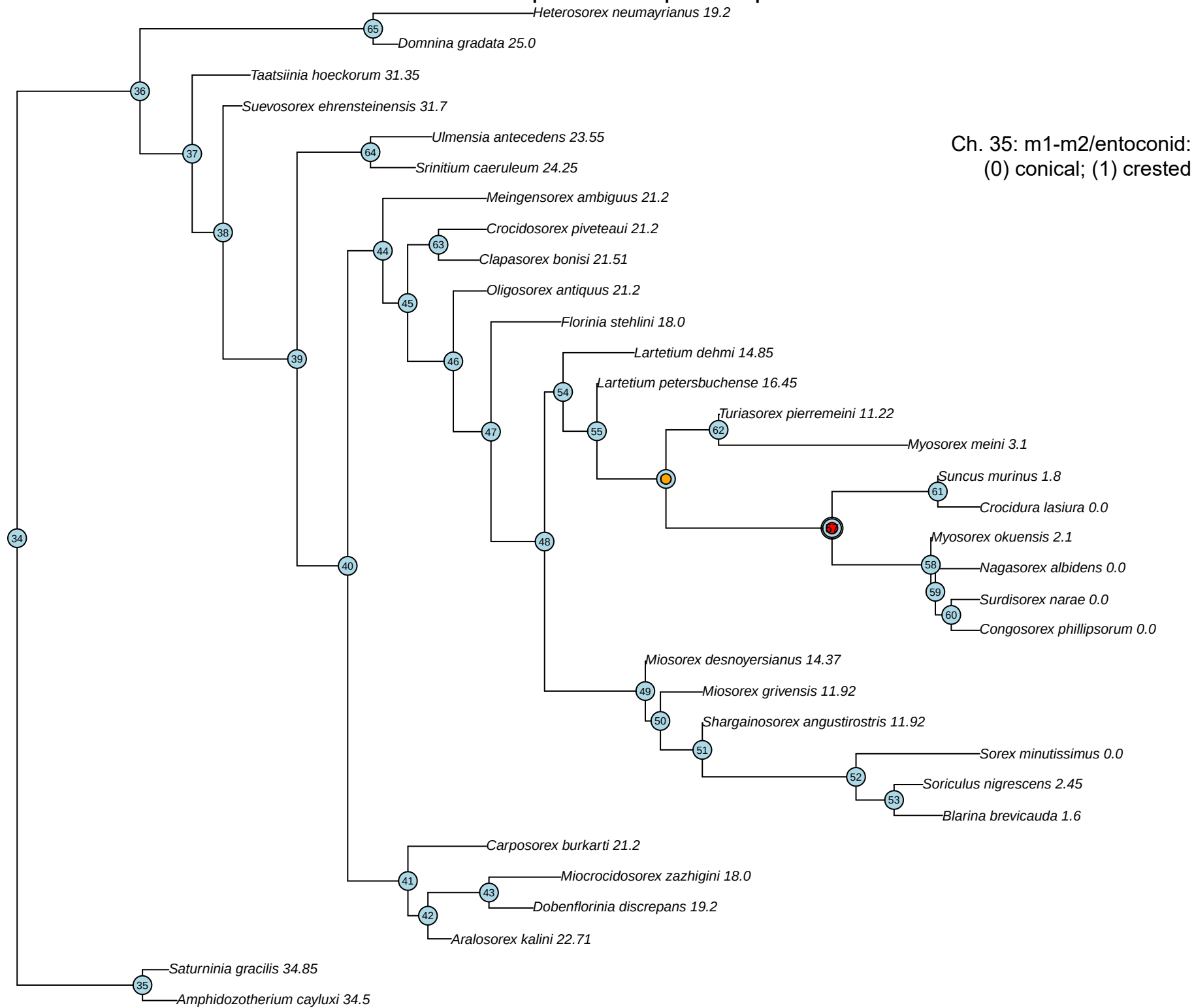
Row 22 | Character 35 | Node 49 | Parent 48



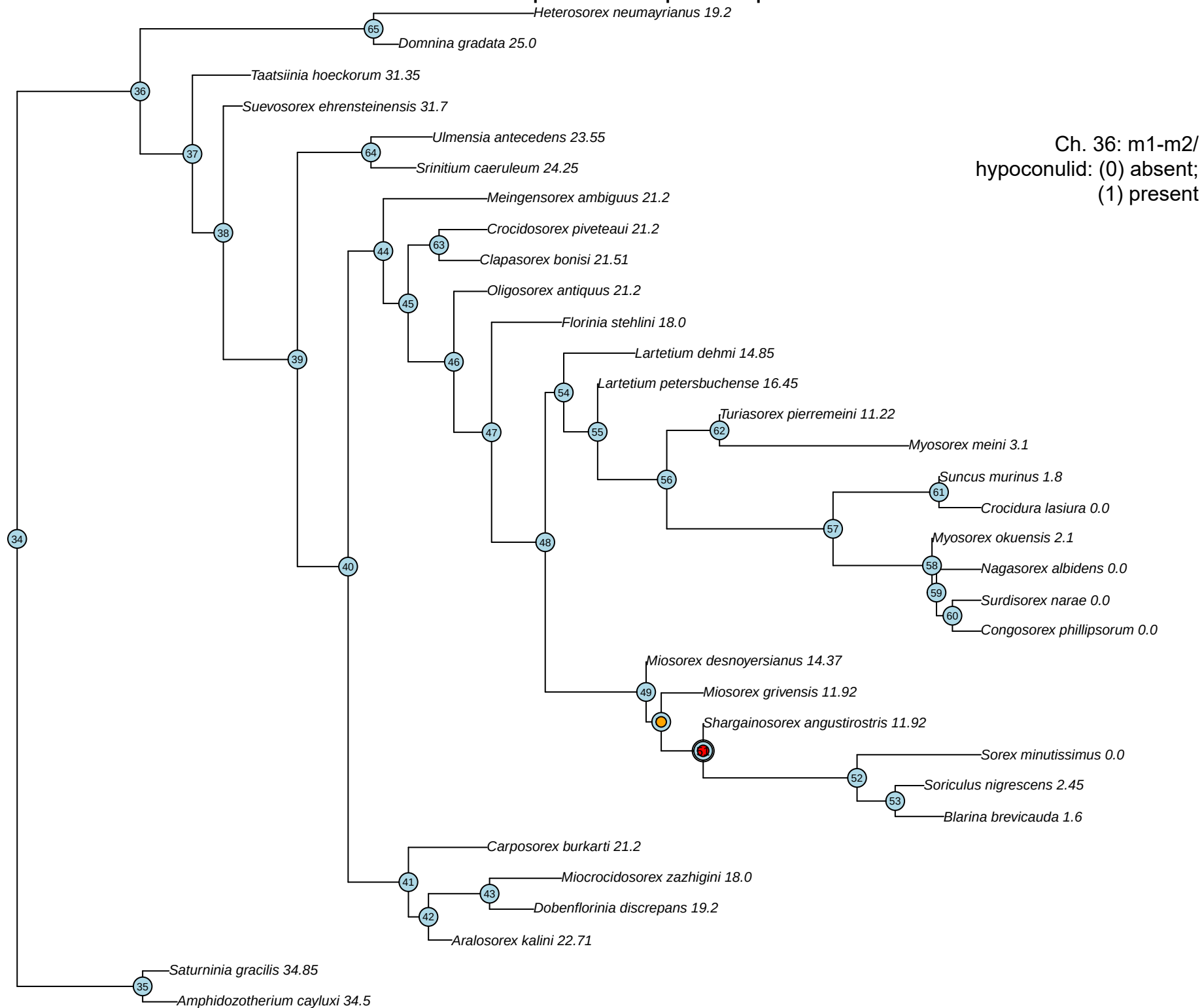
Row 23 | Character 35 | Node 65 | Parent 36



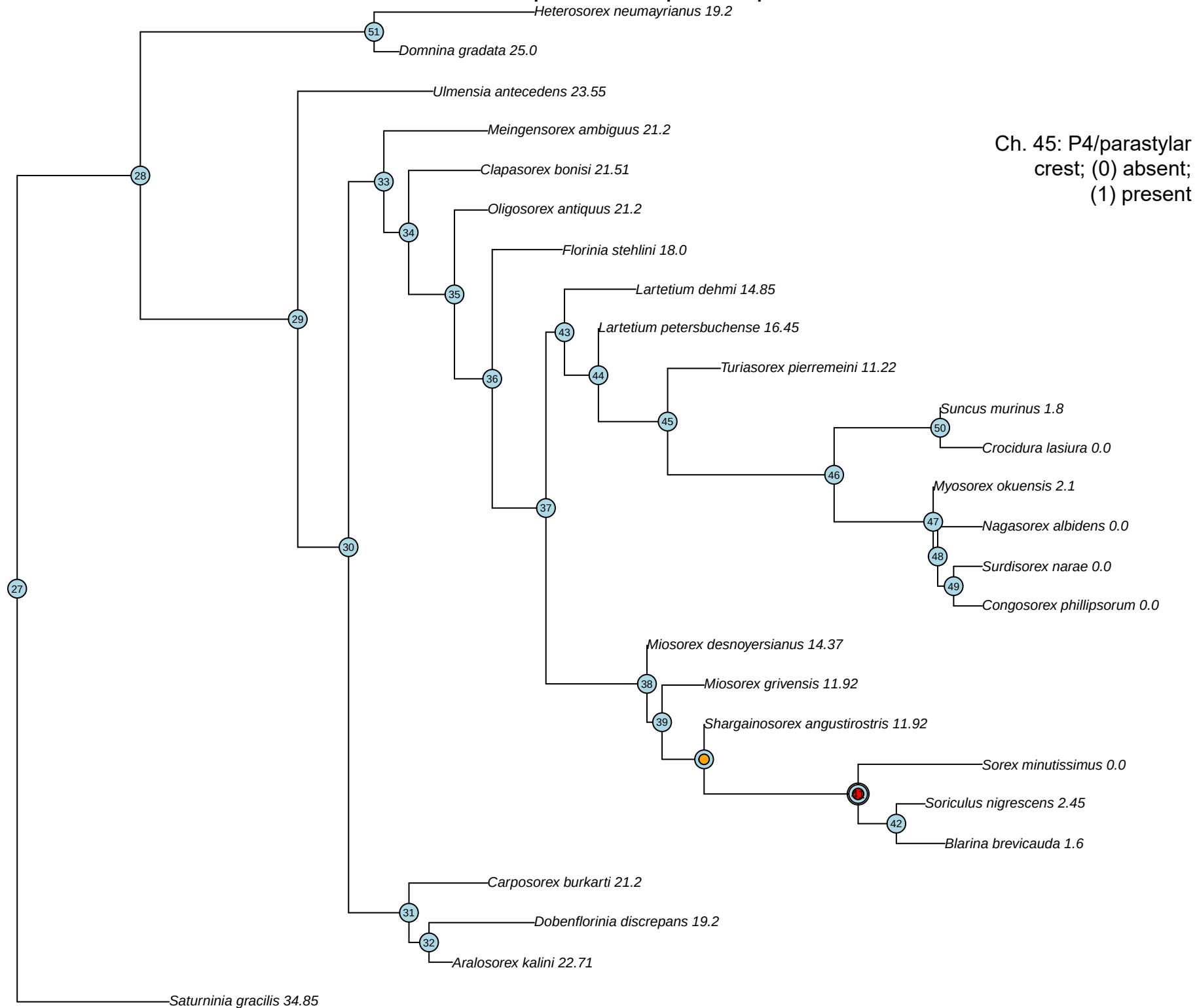
Row 24 | Character 35 | Node 57 | Parent 56



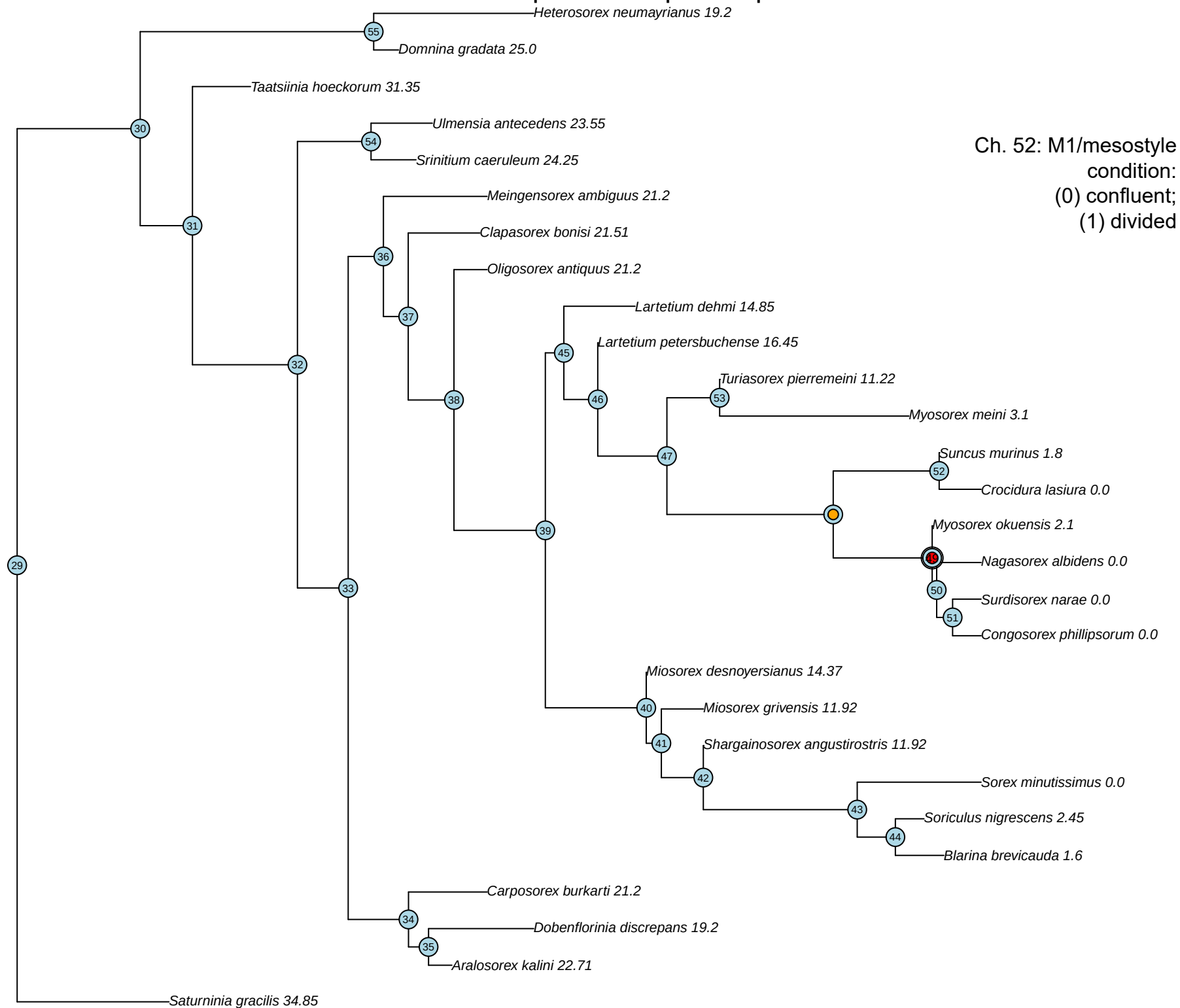
Row 25 | Character 36 | Node 51 | Parent 50



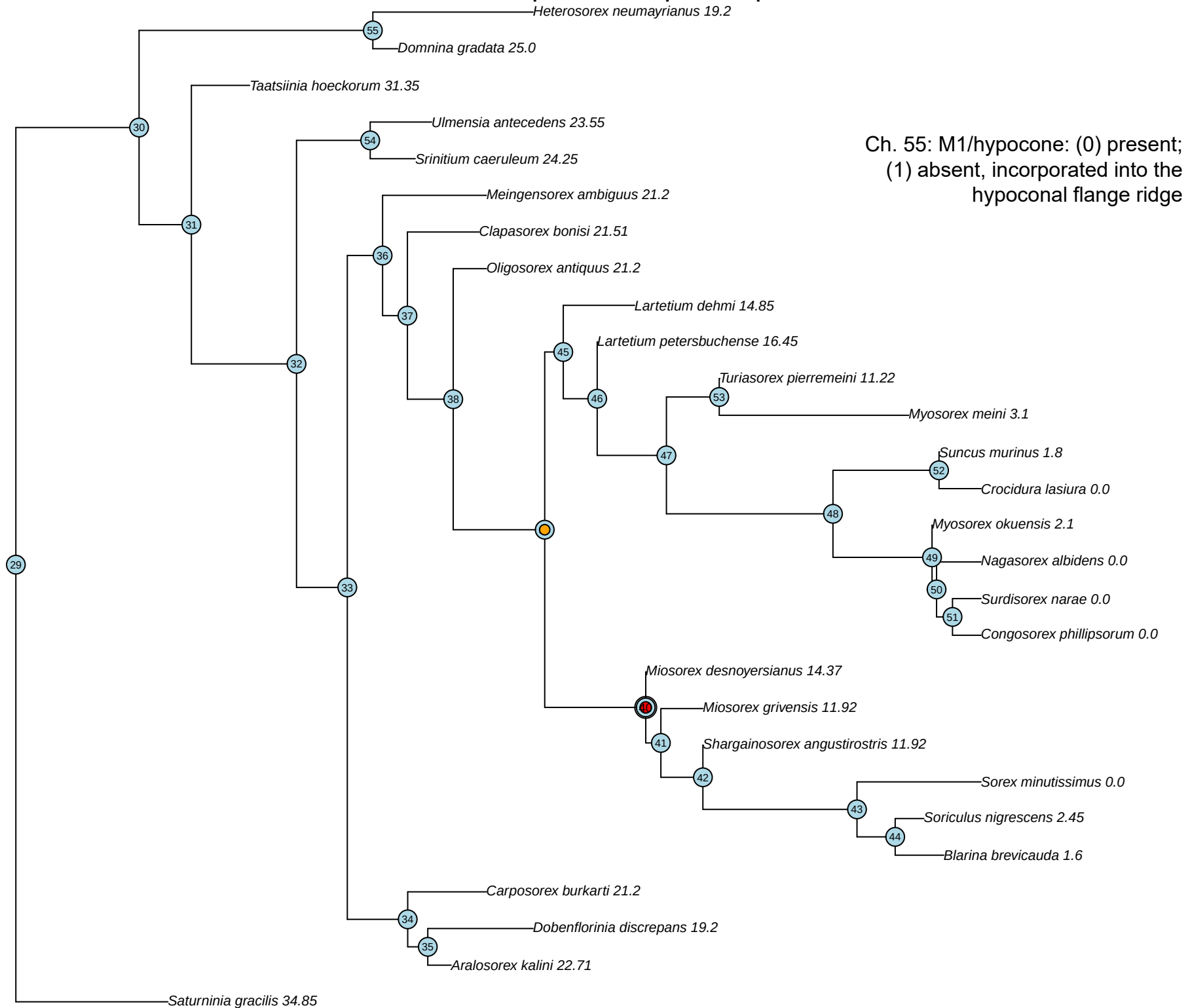
Row 26 | Character 45 | Node 41 | Parent 40



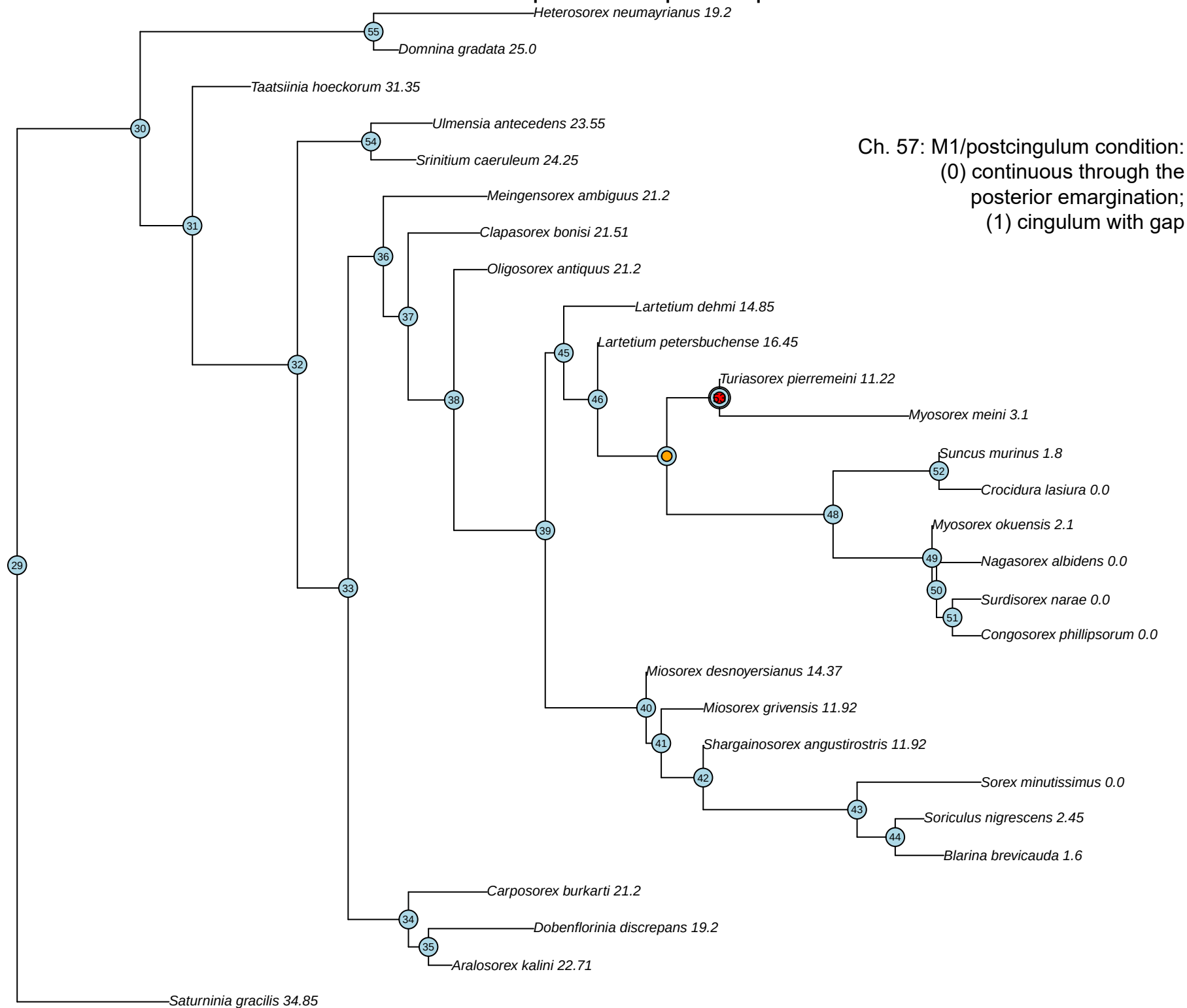
Row 27 | Character 52 | Node 49 | Parent 48



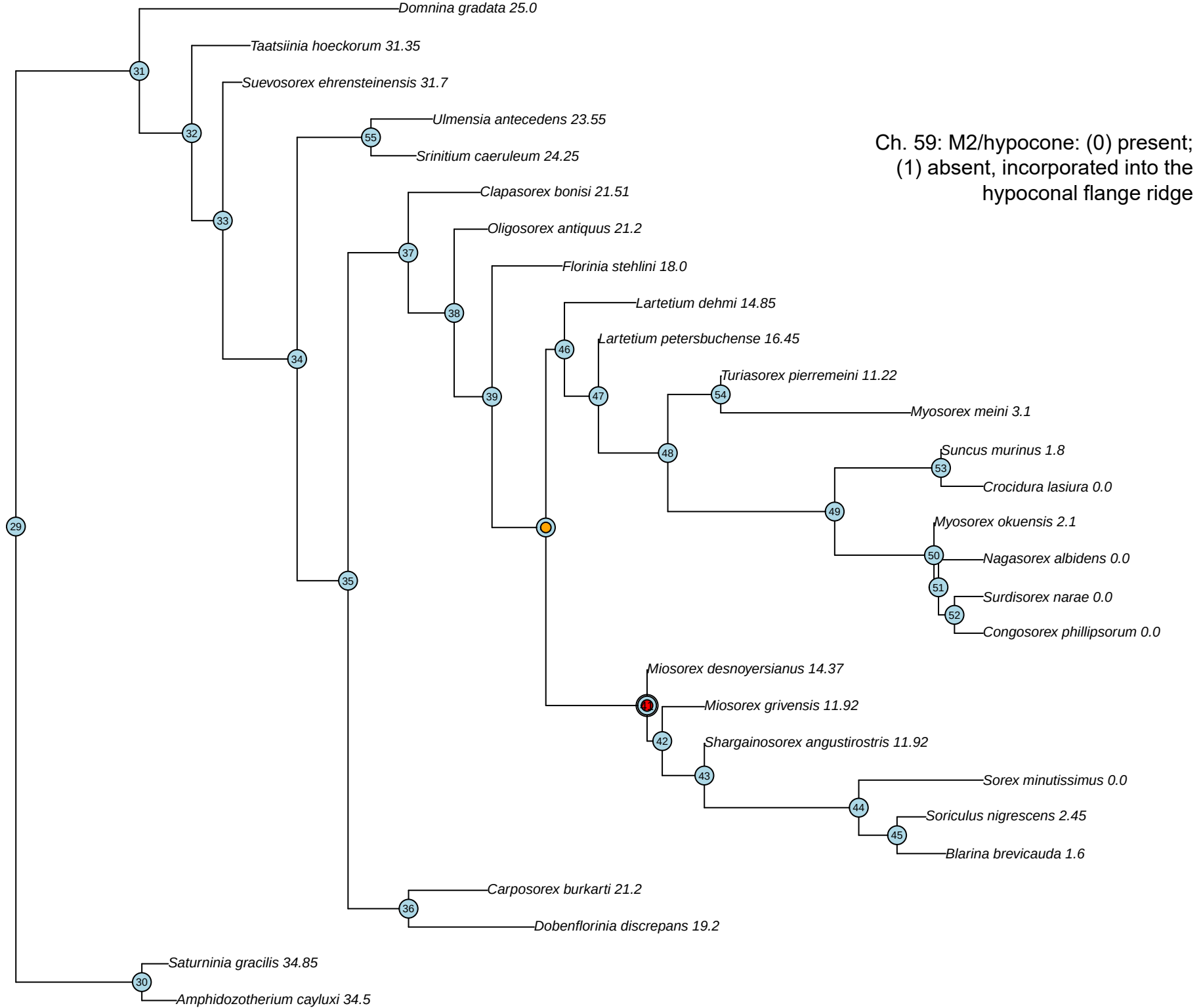
Row 28 | Character 55 | Node 40 | Parent 39



Row 29 | Character 57 | Node 53 | Parent 47



Row 30 | Character 59 | Node 41 | Parent 40



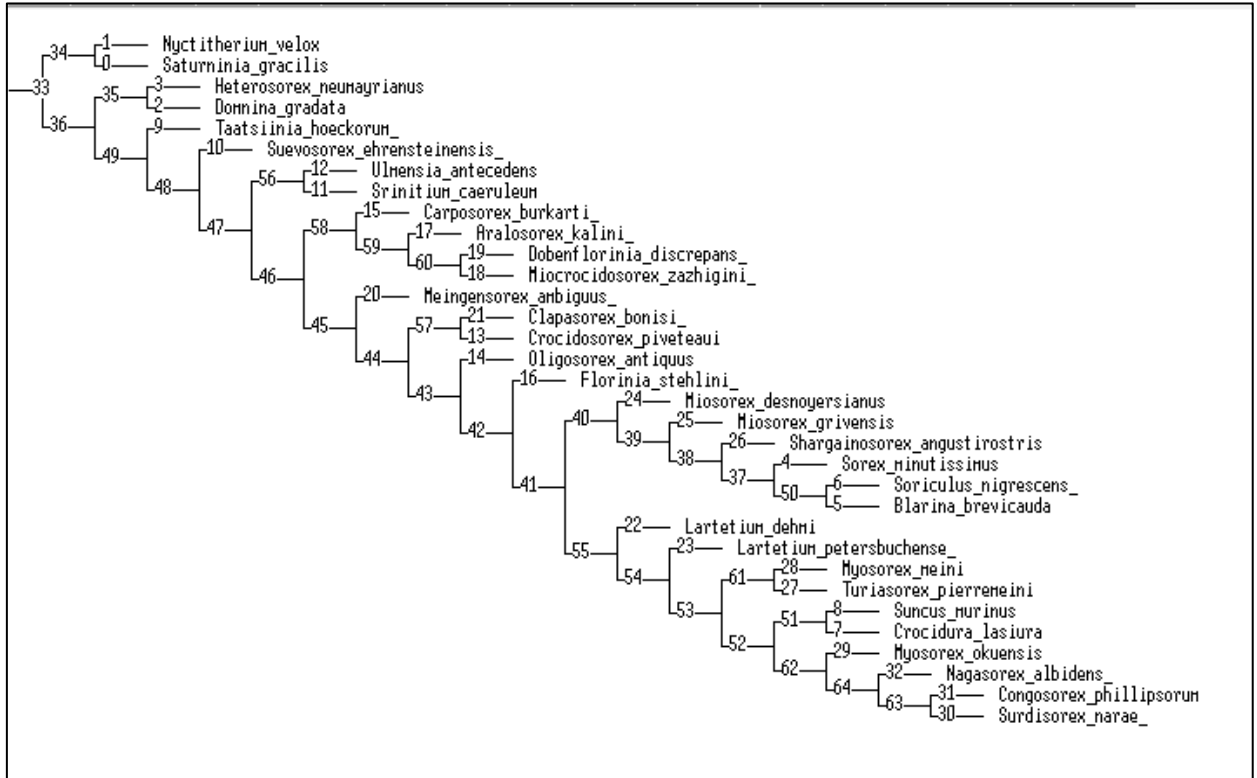


Fig. S10.2. Time-calibrated MCC tree with node labels (TNT)

Supplementary materials: Table S10.2. Synapomorphy list obtained via TNT mapping on the full MCC tree.

(Node numbers refer see above; in TNT character numbering begins at 0)

Saturninia_gracilis :

All trees:

No autapomorphies:

Nyctitherium_velox :

All trees:

m1-m2/ectocingulid (30): (0)_absent --> (1)_discontinuous

Domnina_gradata :

All trees:

Mandibular_condyle/Proportion (12): (2)_height_slightly_exceeds_width -->

(1)_width_slightly_exceeds_height

m1-m2/oblique_cristid (33): (1)_runs_and_terminates_centrally_below_pn -->
 (0)_oblique_cristid_runs_lingually_of_pn
 m3/talonid_tubercles (36): (0)_two_cuspids_[hypoconid,_entoconid] -->
 (1)_single_cuspid_[hypoconid]

Heterosorex_neumayrianus :

All trees:

il_crown/Shape (16): (0)_short --> (1)_elongated
 m3/talonid_cristids (37): (0)_oblique_cristid_and_entocristid_are_developed -->
 (1)_oblique_cristid_is_developed,_but_the_entoconid_is_reduced
 m3/talonid_basin (38): (1)_closed_ --> (0)_open
 I1/apex_tip_condition (39): (0)_non-fissident --> (1)_fissident

Sorex_minutissimus :

All trees:

Mental_foramen/Depression (6): (1)_present --> (0)_absent
 il_crown/Cusps (17): (2)_bicusculate --> (3)_tricusculate
 P4/postcingulum_condition (46): (0)_continuous_through_the_posterior_emargination -->
 (1)_cingulum_with_gap

Blarina_brevicauda :

All trees:

m1-m2/protoconid (28): (1)_the_fold_does_not_reach_the_cingulid_[high_hypoflexid] -->
 (0)_distolabial_fold_reaches_the_ectocingulid_[low_hypoflexid]
 m3/talonid_tubercles (36): (0)_two_cuspids_[hypoconid,_entoconid] -->
 (1)_single_cuspid_[hypoconid]
 P4/ectocingulum_development (45): (1)_present_in_anterior_half_of_crown -->
 (0)_weak_or_absent
 P4/posterior_emargination (47): (0)_deep --> (1)_shallow
 P4/hypocone (50): (1)_absent,_incorporated_into_the_hypoconal_flange_ridge_(cingulum --
 > (0)_present
 M1/mesostyle_condition (51): (0)_confluent --> (1)_divided
 M1/metaloph (53): (0)_absent --> (1)_present
 M2/mesostyle_condition (57): (0)_confluent --> (1)_divided

Soriculus nigrescens :

All trees:

Mental_foramen/Position (5): (4)_below_trigonid_of_m1 --> (5)_distally_trigonid_of_m1

i1_crown/Cusps (17): (2)_bicusperate --> (1)_unicusperate

Antemolars_before_P4 (42): (0)_five --> (1)_four

P4/parastyle_development (43): (0)_weak_or_moderate, _parastylar_lobe_short -->
(1)_strongly_projected_anteriorly, _parastylar_lobe_long, _the_protocone_lies_posteriorly_

M1/hypocone_shape (55): (0)_bulge-like --> (1)_crest-like

M2/hypocone_shape (59): (0)_bulge-like --> (1)_crest-like

Crocidura lasiura :

All trees:

Lower_condylar_facet/Inclination (14): (1)_inferiolingual_inclination_of_the_axis -->
(0)_longitudinal_axis_of_the_facet_is_subhorizontal

p4_mesial_root (26): (0)_well-developed --> (1)_weak_or_absent

Antemolars_before_P4 (42): (1)_four --> (2)_three

Suncus murinus :

All trees:

Lower_condylar_facet/Level (13): (0)_above_the_level_of_the_lower_molars_base -->
(1)_at_the_level_of_the_molar_base

m3/talonid_cristids (37): (0)_oblique_cristid_and_entocristid_are_developed -->

(1)_oblique_cristid_is_developed, _but_the_entoconid_is_reduced

m3/talonid_basin (38): (1)_closed_ --> (0)_open

Taatsiinia hoeckorum :

All trees:

No autapomorphies:

Suevosorex ehrensteinensis :

All trees:

M2/mesostyle_condition (57): (0)_confluent --> (1)_divided

Srinitium_caeruleum :

All trees:

Mental_foramen/Position (5): (2)_below_anterior_root_of_p4 -->

(1)_close_mesially_from_p4

m1-m2/oblique_cristid (33): (0)_oblique_cristid_runs_lingually_of_pn -->

(1)_runs_and_terminates_centrally_below_pn

Ulmensia_antecedens :

All trees:

No autapomorphies:

Crocidosorex_piveteaui :

All trees:

Morphology_of_p4 (23): (2)_two_sub-equal_crests -->

(3)_labial_crest_longer_than_lingual_crest

p4_posterior_sulcus (25): (0)_present --> (1)_absence

m1-m2/ectocingulid (30): (2)_continuous --> (0)_absent

m1-m2/postentoconid_valley (32): (1)_free-ended_hypolophid_with_narrow_valley -->

(2)_free-ended_hypolophid_with_wide_valley

Oligosorex_antiquus :

All trees:

Mandibular_foramen (8): (0)_below_the_middle_part_of_the_internal_temporal_fossa -->

(1)_shifted_at_the_level_of_the_posterior_corner_of_the_fossa

Lower_condylar_facet/Inclination (14): (1)_inferiolingual_inclination_of_the_axis -->

(0)_longitudinal_axis_of_the_facet_is_subhorizontal

m1-m2/ectocingulid (30): (2)_continuous --> (1)_discontinuous

Carposorex_burkarti :

All trees:

Lower_condylar_facet/Inclination (14): (1)_inferiolingual_inclination_of_the_axis -->

(0)_longitudinal_axis_of_the_facet_is_subhorizontal

il_crown/Cusps (17): (2)_bicusperate --> (1)_unicusperate

m3/talonid_cristids (37): (0)_oblique_cristid_and_entocristid_are_developed -->
 (1)_oblique_cristid_is_developed,_but_the_entoconid_is_reduced
 m3/talonid_basin (38): (1)_closed_ --> (0)_open
 M1/postmetacrista (52): (1)_longer_than_postparacrista --> (0)_subequal_to_postparacrista

Florinia_stehlini_ :

All trees:

Mental_foramen/Position (5): (3)_below_posterior_root_of_p4 -->
 (4)_below_trigonid_of_m1
 m3/talonid_cristids (37): (0)_oblique_cristid_and_entocristid_are_developed -->
 (1)_oblique_cristid_is_developed,_but_the_entoconid_is_reduced
 m3/talonid_basin (38): (1)_closed_ --> (0)_open
 P4/posterior_emargination (47): (0)_deep --> (1)_shallow
 P4/protocone_development (48): (0)_poor,_with_the_short_postprotocrista -->
 (2)_protocone_undeveloped_as_a_cusp

Aralosorex_kalini_ :

All trees:

No autapomorphies:

Miocrocidosorex_zazhigini_ :

All trees:

Small_antemolar_is_incrusted_under_p4 (22):
 (0)_absent,_a_last_antemolar_is_moderate_developed --> (1)_present_as_a_vestigial_tooth
 m1-m2/ectocingulid (30): (2)_continuous --> (1)_discontinuous
 m3/talonid_tubercles (36): (0)_two_cuspids_[hypoconid,_entoconid] -->
 (1)_single_cuspid_[hypoconid]

Dobenflorinia_discrepans_ :

All trees:

m1-m2/protoconid (28): (0)_distolabial_fold_reaches_the_ectocingulid_[low_hypoflexid] --
 > (1)_the_fold_does_not_reach_the_cingulid_[high_hypoflexid]

Meingensorex_ambiguus_ :

All trees:

i1_crown/Cusps (17): (2)_bicusculate --> (1)_unicusculate

Antemolars_before_p4 (21): (2)_three --> (3)_two

m1-m2/ectocingulid (30): (2)_continuous --> (1)_discontinuous

m3/talonid_cristids (37): (0)_oblique_cristid_and_entocristid_are_developed -->

(1)_oblique_cristid_is_developed,_but_the_entoconid_is_reduced

m3/talonid_basin (38): (1)_closed_ --> (0)_open

P4/protocone_development (48): (0)_poor,_with_the_short_postprotocrista -->

(2)_protocone_undeveloped_as_a_cusp

M1/hypocone_shape (55): (0)_bulge-like --> (1)_crest-like

Clapasorex_bonisi_ :

All trees:

Mental_foramen/Position (5): (3)_below_posterior_root_of_p4 -->

(2)_below_anterior_root_of_p4

Antemolars_before_p4 (21): (2)_three --> (1)_four

Small_antemolar_is_incrusted_under_p4 (22): (1)_present_as_a_vestigial_tooth -->

(0)_absent,_a_last_antemolar_is_moderate_developed

m3/talonid_tubercles (36): (0)_two_cuspids_[hypoconid,_entoconid] -->

(1)_single_cuspid_[hypoconid]

Lartetium_dehmi_ :

All trees:

P4/parastyle_development (43): (0)_weak_or_moderate,_parastylar_lobe_short -->

(1)_strongly_projected_anteriorly,_parastylar_lobe_long,_the_protocone_lies_posteriorly_

Lartetium_petersbuchense_ :

All trees:

m1-m2/ectocingulid (30): (2)_continuous --> (1)_discontinuous

Miosorex_desnoyersianus_ :

All trees:

Postsymphyseal_foramen (7): (3)_below_m1 --> (2)_below_p4

Miosorex_grivensis :

All trees:

m1-m2/size (27): (1)_m2_shorter_than_m1 --> (0)_m2_is_subequal_to_m1_in_length

P4/ectocingulum_development (45): (1)_present_in_anterior_half_of_crown, or (2)_well-developed --> (0)_weak_or_absent

Shargainosorex_angustirostris :

All trees:

m1-m2/labial_notch (29): (0)_present --> (1)_absent

Turiasorex_pierremeini :

All trees:

Mental_foramen/Position (5): (3)_below_posterior_root_of_p4 -->
(4)_below_trigonid_of_m1

Myosorex_meini :

All trees:

m1-m2/size (27): (1)_m2_shorter_than_m1 --> (0)_m2_is_subequal_to_m1_in_length

Myosorex_okuensis :

All trees:

P4/protocone_development (48): (0)_poor,_with_the_short_postprotocrista -->
(1)_complex,_with_the_relatively_long_postprotocrista_(in_some_taxa_+_metaloph)

Surdisorex_narae_ :

All trees:

Antemolars_before_p4 (21): (3)_two --> (4)_one

Congosorex_phillipsorum :

All trees:

Mental_foramen/Position (5): (3)_below_posterior_root_of_p4 -->
(4)_below_trigonid_of_m1

Nagasurex_albidens_ :

All trees:

Mental_foramen/Depression (6): (1)_present --> (0)_absent

m1-m2/ectocingulid (30): (2)_continuous --> (1)_discontinuous

Antemolars_before_P4 (42): (1)_four --> (0)_five

Node 34 :

All trees:

No synapomorphies

Node 35 :

All trees:

Mental_foramen/Depression (6): (0)_absent --> (1)_present

Lower_condylar_facet/Level (13): (0)_above_the_level_of_the_lower_molars_base -->

(1)_at_the_level_of_the_molar_base

i1_crown/Cusps (17): (2)_bicusperate --> (0)_acusperate

m1-m2/entoconid (34): (0)_conical --> (1)_crested

P4/postcingulum_condition (46): (1)_cingulum_with_gap -->

(0)_continuous_through_the_posterior_emargination

Node 36 :

All trees:

No synapomorphies

Node 37 :

All trees:

Mandibular_condyle/Size (4): (1)_intermediate --> (2)_large

Mandibular_foramen (8): (0)_below_the_middle_part_of_the_internal_temporal_fossa -->

(1)_shifted_at_the_level_of_the_posterior_corner_of_the_fossa

Mandibular_condyle/Shape (11):

(2)_divided_with_lingual_fusion_of_the_upper_and_lower_facets_[crocidurine-like] -->

(3)_upper_and_lower_facets_clearly_defined_[soricine-like]

p4_posterior_sulcus (25): (0)_present --> (1)_absence

P4/parastylar_crest (44): (0)_absent --> (1)_present

Node 38 (numbered as node 51 on the corresponding MCC tree in the main text) :

All trees:

Mental_foramen/Position (5): (3)_below_posterior_root_of_p4 -->

(4)_below_trigonid_of_m1

Antemolars_before_p4 (21): (3)_two --> (4)_one

Small_antemolar_is_incrusted_under_p4 (22): (1)_present_as_a_vestigial_tooth -->

(0)_absent, a_last_antemolar_is_moderate_developed

m1-m2/hypoconulid (35): (0)_absent --> (1)_present

Node 39 :

All trees:

m3/talonid_cristids (37): (0)_oblique_cristid_and_entocristid_are_developed -->

(1)_oblique_cristid_is_developed, but_the_entoconid_is_reduced

m3/talonid_basin (38): (1)_closed_ --> (0)_open

Node 40 :

All trees:

Mandibular_condyle/Proportion (12): (2)_height_slightly_exceeds_width -->

(3)_height_significantly_exceeds_width

m1-m2/postentoconid_valley (32): (1)_free-ended_hypolophid_with_narrow_valley -->

(2)_free-ended_hypolophid_with_wide_valley

M1/hypocone (54): (0)_present -->

(1)_absent, incorporated_into_the_hypoconal_flange_ridge

M2/hypocone (58): (0)_present -->

(1)_absent, incorporated_into_the_hypoconal_flange_ridge

Node 41 (numbered as node 48 on the corresponding MCC tree in the main text) :

All trees:

il_crown-root_axis (18): (0)_crown_and_root_axes_form_an_angle --> (1)_axes_aligned

Node 42 :

All trees:

il_crown/Shape (16): (0)_short --> (1)_elongated

i1_labial_extension (19): (0)_labial_and_lingual_side_are_equal_in_length -->
(1)_labial_side_is_extended

Node 43 (numbered as node 46 on the corresponding MCC tree in the main text) :

All trees:

Mental_foramen/Position (5): (2)_below_anterior_root_of_p4 -->
(3)_below_posterior_root_of_p4
Small_antemolar_is_incrusted_under_p4 (22):
(0)_absent, a_last_antemolar_is_moderate_developed --> (1)_present_as_a_vestigial_tooth
m1-m2/protoconid (28): (0)_distolabial_fold_reaches_the_ectocingulid_[low_hypoflexid] --
> (1)_the_fold_does_not_reach_the_cingulid_[high_hypoflexid]
Antemolars_before_P4 (42): (1)_four --> (0)_five
P4/parastylar_crest (44): (0)_absent --> (1)_present
P4/postcingulum_condition (46): (1)_cingulum_with_gap -->
(0)_continuous_through_the_posterior_emargination

Node 44 :

All trees:

p4_crown_symmetry_in_occlusal_view (24):
(0)_lingual_and_labial_sides_appear_symmetrical --> (1)_buccal_part_extends_posteriorly

Node 45 :

All trees:

m1-m2/oblique_cristid (33): (1)_runs_and_terminates_centrally_below_pn -->
(0)_oblique_cristid_runs_lingually_of_pn

Node 46 :

All trees:

Zygomatic_arch (0): (0)_present --> (1)_absent
Internal_temporal_fossa (9): (1)_shallow --> (2)_deep
m1-m2/ectocingulid (30): (0)_absent --> (2)_continuous

Node 47 (numbered as node 39 on the corresponding MCC tree in the main text) :

All trees:

m1-m2/postentoconid_valley (32): (2)_free-ended_hypolophid_with_wide_valley -->
 (1)_free-ended_hypolophid_with_narrow_valley
 m3/talonid_cristids (37): (1)_oblique_cristid_is_developed,_but_the_entoconid_is_reduced -
 --> (0)_oblique_cristid_and_entocristid_are_developed
 m3/talonid_basin (38): (0)_open --> (1)_closed_
 P4/protocone_development (48): (0)_poor,_with_the_short_postprotocrista -->
 (1)_complex,_with_the_relatively_long_postprotocrista_(in_some_taxa+_metaloph)
 M1/hypocone (54): (1)_absent,_incorporated_into_the_hypoconal_flange_ridge -->
 (0)_present
 M2/hypocone (58): (1)_absent,_incorporated_into_the_hypoconal_flange_ridge -->
 (0)_present

Node 48 (numbered as node 38 on the corresponding MCC tree in the main text):

All trees:

Antemolars_before_p4 (21): (3)_two --> (4)_one
 I1/medial_groove_of_root (40): (0)_absent --> (1)_present
 P4/postcingulum_condition (46): (0)_continuous_through_the_posterior_emargination -->
 (1)_cingulum_with_gap
 P4/hypocone (50): (0)_present -->
 (1)_absent,_incorporated_into_the_hypoconal_flange_ridge_(cingulum
 M1/hypocone (54): (0)_present -->
 (1)_absent,_incorporated_into_the_hypoconal_flange_ridge
 M2/hypocone (58): (0)_present -->
 (1)_absent,_incorporated_into_the_hypoconal_flange_ridge

Node 49 (numbered as node 37 on the corresponding MCC tree in the main text):

All trees:

m1-m2/entoconid (34): (0)_conical --> (1)_crested

Node 50 :

All trees:

M1/hypocone_shape (55): (0)_bulge-like --> (1)_crest-like
 M2/hypocone_shape (59): (0)_bulge-like --> (1)_crest-like

Node 51 :

All trees:

Antemolars_before_P4 (42): (0)_five --> (1)_four

P4/preprotocrista (49): (1)_riches_the_parastyle --> (0)_short,_not_rich_the_parastyle

Node 52 (numbered as node 57 on the corresponding MCC tree in the main text):

All trees:

P4/hypocone (50): (1)_absent,_incorporated_into_the_hypoconal_flange_ridge_(cingulum --> (0)_present

Node 53 :

All trees:

m1-m2/labial_notch (29): (0)_present --> (1)_absent

Node 54 (numbered as node 55 on the corresponding MCC tree in the main text):

All trees:

Mental_foramen/Position (5): (2)_below_anterior_root_of_p4 --> (4)_below_trigonid_of_m1

Node 55 (numbered as node 54 on the corresponding MCC tree in the main text):

All trees:

m1-m2/postentoconid_valley (32): (1)_free-ended_hypolophid_with_narrow_valley --> (0)_hypolophid_connects_entoconid_[valley_absent]

m3/talonid_cristids (37): (0)_oblique_cristid_and_entocristid_are_developed --> (2)_oblique_cristid_is_shortened_and_the_entoconid_is_lost

m3/talonid_basin (38): (1)_closed_ --> (0)_open

M1/postcingulum_condition (56): (0)_continuous_through_the_posterior_emargination --> (1)_cingulum_with_gap

Node 56 :

All trees:

M1/mesostyle_condition (51): (0)_confluent --> (1)_divided

M2/mesostyle_condition (57): (0)_confluent --> (1)_divided

Node 57 :

All trees:

Antemolars_before_P4 (42): (1)_four --> (2)_three